

MED 419 NEUROSCIENCE II PAST QUESTIONS COMPILATION 2026

INTERNAL MEDICINE

1. The upper motor neuron impairment produces the following change of muscles tone:
 - a. "cog wheel" rigidity
 - b. flaccidity
 - c. myoclonia
 - d. spasticity

2. The muscular wasting (hypotrophy) usually develops with disease in:
 - a. caudate
 - b. cerebellar
 - c. lower motor neuron
 - d. upper motor neuron

3. The temperature & pin sense loss usually develops with disease in:
 - a. anterior horns of spinal cord
 - b. lateral horns of spinal cord
 - c. posterior horns of spinal cord
 - d. posterior columns of spinal cord

4. A glove-&-stocking pattern of sensory disturbance usually develops with disease in:
 - a. peripheral nerves
 - b. the thalamus
 - c. the spinal cord

d. the brainstem

5. Babinsky response usually develops with damage in:

- a. lower motor neuron
- b. upper motor neuron
- c. thalamus
- d. cerebellar

6. The presence of ataxia suggests damage to any of the following EXCEPT:

- a. cerebellar
- b. thalamus
- c. vestibular nucleus
- d. vagal nerve

7. The ability to walk along a straight line is most often impaired with:

- a. cerebellar dysfunction
- b. parietal lobe damage
- c. temporal lobe damage
- d. ocular motor disturbances

8. Parkinsonism includes combination of the following:

- a. tremor, bradykinesia & muscles rigidity
- b. paresis, anesthesia & muscles spasticity
- c. chorea & muscles hypotonia
- d. tremor, ataxia & muscles hypotonia

9. Hemiplegia, hemianesthesia & hemianopia develop together with disease in the:

- a. spinal cord
- b. internal capsule
- c. thalamus
- d. brainstem

10. Affection of the cerebellar may produce any of the following EXCEPT:

- a. positive Romberg's test
- b. positive finger to nose test
- c. positive heel to knee test
- d. positive Rinner & Weber test

11. Brown-Sequard syndrome develops with the following damage of the spinal cord:

- a. complete transversal
- b. anterior horns
- c. half transversal
- d. posterior horns

12. The presence of ptosis suggests damage to cranial nerve:

- a. IV
- b. V
- c. III
- d. VII

13. The presence of dysphagia suggests damage to cranial nerves:

- a. V-VII
- b. IX-X
- c. VII-XI

d. III-VI

14. The presence of dysarthria suggests damage to cranial nerve:

a. V

b. XI

c. XII

d. VIII

15. The damage to IX, X & XII cranial nerves produce:

a. bulbar palsy

b. pseudobulbar palsy

c. Brown-Sequard syndrome

d. Argyle-Robertson syndrome

16. Dysphasia suggests the impairment of:

a. speech

b. gait

c. swallowing

d. movement

17. The Broca's area is located in the lobe:

a. frontal

b. parietal

c. temporal

d. occipital

18. The patient with apraxia cannot:

- a. name his fingers
- b. carry out an imagined act
- c. draw simple diagrams
- d. speak fluently

19. Meningeal sign is the following:

- a. Babinsky
- b. Kernig
- c. Lasseg
- d. Romberg

20. Any of the following syndromes is the involuntary movement EXCEPT:

- a. chorea
- b. tic
- c. tremor
- d. paresis

21. Pathological reflex, occurred in central paralysis (upper motor neuron lesion) is the following:

- a. Brudzinsky
- b. Nery
- c. Babynsky
- d. Brown-Sequard

22. Fibrillations (fasciculations) may develop with disease in:

- a. lateral column of the spinal cord
- b. posterior horn of the spinal cord

- c. anterior horn of the spinal cord
- d. internal capsule

23. Bilateral affection of spinal cord at the cervical level may produce the following syndrome:

- a. hemiplegia
- b. paraplegia
- c. tetraplegia
- d. monoplegia

24. Central paresis, loss of proprioceptive sensation on one side & loss of exteroceptive sensation on the opposite form the following syndrome:

- a. Lambert-Iton
- b. Matskevich-Shtrumpel
- c. Argile-Robertson
- d. Brown-Sequard

25. The polyneuropathic pattern of sensory loss suggests presence of the following syndrome:

- a. numbness & pain in distal parts of extremities
- b. numbness & analgesia in half of the body
- c. pain & sensory ataxia in half of the body
- d. analgesia & sensory ataxia in proximal parts of extremities

26. The presence of hemianesthesia, hemianopia & sensory hemiataxia suggests damage to the following:

- a. internal capsule
- b. thalamus opticus
- c. spinal cord
- d. black substance

27. The presence of Lasegue sign suggests damage to the following:

- a. meninges of the brain
- b. spinal roots C5-C8 or radial nerve
- c. spinal roots L5-S1 or sciatic nerve
- d. anterior horns at the level L5-S1

28. The affection of cerebellar may produce any of the following EXCEPT:

- a. nystagmus
- b. ataxia
- c. dysmetria
- d. dyspraxia

29. The presence of dysdiadochokinesis suggests damage to the following:

- a. black substance
- b. spinal cord
- c. cerebellar
- d. occipital lobe

30. The presence of Parkinsonism suggests damage to the following:

- a. caudate nucleus
- b. black substance
- c. cerebellar
- d. frontal lobe

31. In initial stage of Parkinson disease the most typical involuntary movement is the following:

- a. chorea

- b. atetosis
- c. tremor
- d. dystonia

32. The autonomic nervous system includes any of the following EXCEPT:

- a. hypothalamus
- b. paravertebral sympathetic trunk
- c. vagal nerve
- d. cerebral cortex

33. One of the most important functions of the autonomic nervous system is the following:

- a. regulation of homeostasis
- b. voluntary movements
- c. coordination of movements
- d. involuntary movements

34. Any of the following cranial nerves has the parasympathetic nucleus EXCEPT:

- a. vagal
- b. oculomotor
- c. glossopharyngeal
- d. olfactory

35. The presence of anosmia suggests damage to the following cranial nerve:

- a. II
- b. I
- c. III
- d. V

36. Trigeminal nerve impairment produces the following symptoms:

- a. plegia in half of the face
- b. ache paroxysm in half of the face
- c. disturbance of swallowing
- d. ache in half of the head

37. The presence of Bell's palsy suggests damage to the following cranial nerve:

- a. facial
- b. optic
- c. olfactory
- d. vestibular

38. Dysphagia, dysphonia, dysarthria together with tongue atrophy & depressed "gag" reflex is called like following:

- a. bulbar palsy
- b. Bell's palsy
- c. pseudobulbar palsy
- d. bulbus olfactorius

39. The disturbance of purposive movement in absence of paresis & dyscoordination suggests the presence of the following:

- a. dyslexia
- b. dysgnosia
- c. dyspraxia
- d. dysphasia

40. Meningeal syndrome suggests any of the following, EXCEPT:

- a. neck stiffness
- b. headache
- c. photophobia
- d. Babynsky response

41. Multiple sclerosis is a demyelinating disorder that results in deficits. It is ultimately fatal. This course of disease is most identical to the outcome of which one of the following demyelinating disorders:

- a. Acute Necrotizing Hemorrhagic Encephalomyelitis
- b. Neuromyelitis Optica
- c. Acute disseminated encephalomyelitis
- d. Central pontine myelinolysis

42. Choose the FALSE statement concerning vasovagal and reflex anoxic seizures:

- a. There is increased risk of epilepsy
- b. Incontinence may be present
- c. Seizure duration of less than 60 seconds
- d. Reassurance is the mainstay of management

43. Which of the follow is characteristic of the Miller — Fischer Variant?

- a. Hyperreflexia
- b. CSF shows marked leukocytosis
- c. Bradykinesia
- d. EMG shows low amplitude sensory and motor evoked responses

44. A 18 year old male complains of symptoms that started 3 years ago with difficulty in walking and frequent tripping due to foot and distal leg weakness which has gradually worsened. Distal muscle wasting is noted in the legs giving the appearance of an inverted wine bottle. He has a high-arched foot and the toes are always curled. Deep tendon reflexes are found to be markedly

diminished. There is a loss of vibration, pain, and temperature senses in the feet. Which of the following would be useful in confirming the diagnosis?

- a. Electroencephalogram
- b. MRI brain
- c. Electromyography and nerve conduction studies
- d. X-ray of the thoracolumbar spine

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- a. Multiple sclerosis
- b. Hereditary motor and sensory neuropathy
- c. Pott's disease
- d. Acute inflammatory demyelinating polyneuropathy

46. The symptoms and signs of multiple sclerosis are not a result of the:

- a. Limited capacity of the CNS to regenerate normal myelin
- b. Effect of myelin loss on the transmission of electrical impulses along axons
- c. Degree of secondary damage to axons that occurs as the disease runs its course
- d. Transection of axons during the course of the disease

47. A 54 year old farmer is brought by his wife into the outpatient clinic with complaints that he has become very sluggish. On examination the attending doctor finds increased tone in the limbs and notices tremors in both hands. The patient face shows little emotion and he walks with a bent posture. What other symptoms might you expect in this patient?

- a. Previous history of diarrhea
- b. Difficulty maintaining balance

- c. Breathlessness
- d. Numbness in the hands and feet

48. The gag reflex done during neurological assessment of a deeply comatose patient tests the viability of:

- a. Whole brain-stem
- b. Mid-brain
- c. Medulla oblongata
- d. Pons

49. The first symptom that most patient's with Parkinson's disease experience is:

- a. Stooped posture.
- b. Resting tremor.
- c. Shuffling gait.
- d. Rigidity.

50. Concerning Gullian Barre Syndrome, which of the following is TRUE:

- a. Weakness usually asymmetrical
- b. CSF shows elevated protein, normal cell count and glucose
- c. There is a rapidly evolving spastic paralysis
- d. It has a relapsing or steadily progressive course over months or years

51. With regards to Cerebral palsy:

- a. Cerebral damage is limited to the motor system
- b. Spastic type is the least common
- c. A lumbar puncture is mandatory to confirm the diagnosis
- d. Seizures are a rare association

52. The best way to initially test for cranial nerve VIII auditory component is:

- a. Do a Weber's test with a tuning fork
- b. Close both ears and whisper into both ears
- c. Close one ear and whisper into the other ear
- d. Do a Rhine's test with a tuning fork

53. A 60 year old lady presents with a month long history of general malaise, headaches and scalp tenderness. A diagnosis of temporal arteritis is suspected. Which of the following drugs is likely to benefit her in the long term?

- a. Prednisolone
- b. Paracetamol
- c. Sumatriptan
- d. Aspirin

54. The treatment of choice for mononeuropathy multiplex is:

- a. High dose cerebrotonin
- b. Intermittent Vitamin B injections
- c. Glucocorticoids and cyclophosphamide
- d. High dose meropenem

55. All of the following are cardinal signs for Parkinson's disease EXCEPT:

- a. Bradykinesia.
- b. Cogwheel rigidity.
- c. Symmetric onset.
- d. Distal resting tremor.

56. 20-year-old patient who sustained a left temporal lobe contusion from a road traffic accident is now reporting with a non-motor seizure. Which seizure type is most likely to be exhibited?

- a. Tonic-clonic seizures
- b. Tonic seizures
- c. Formications
- d. Hallicinations

57. The brain is ----- of body mass yet uses ----- of the daily glucose requirement of the body:

- a. 10%, 20%
- b. 20%, 10%
- c. 5%, 20%
- d. 15%, 15%

58. A 55 year old male complains of severe headaches that started three weeks ago. He describes the pain as throbbing and localized to the temporal area. He complains of occasional low-grade fevers. He also admits pain when combing his hair and when chewing food. He is experiencing some blurred vision in his right eye. Possible complications of this patient's condition include:

- a. Stroke
- b. Renal failure
- c. Hyperuricemia
- d. Thrombophilia

59. An amateur boxer who is confirmed to have suffered several episodes of concussion after many years is now found to have progressively worsening neurological status including bowel and urinary incontinence. What complication is he most likely to be suffering from:

- a. Chronic concussion syndrome
- b. Punch drunk syndrome
- c. Chronic punching syndrome

d. Acquired motor neurone disease

60. A 45 year old male patient is seen with a claw-hand deformity most pronounced in 4th & 5th digits. Further examination reveals sensory loss occurs over the fifth finger, the ulnar aspect of the fourth finger, and the ulnar border of the palm. Which of the following is the most likely pathophysiology underlying the patient's presentation?

- a. Antigenic mimicry and demyelination of nerves
- b. Destruction of acetylcholine receptors at the neuromuscular end plate.
- c. Cerebrovascular accident
- d. Nerve compression

61. Which of the following conditions is a cause of compressive peripheral neuropathies?

- a. Cytomegalovirus infection
- b. Infectious mononeucleosis
- c. Pregnancy
- d. HIV infection

62. A 34-year-old woman presents with blurry and dim vision in the left eye and some paresthesia, especially involving the right upper limb. She says that she has had occasional bouts of poor vision over the past few years as well as occasional muscle weakness. What disorder is she most likely experiencing?

- a. Guillian Barre syndrome
- b. Amyotrophic lateral sclerosis
- c. Duchenne muscular dystrophy
- d. Multiple sclerosis

63. Which of the following abnormalities would NOT be expected in a severe, ulnar neuropathy?

- a. brachioradialis reflexes are decreased
- b. Positive Froment sign

- c. numbness in the fifth digit
- d. claw hand

64. A 67-year-old woman said she was unable to use utensils to eat her food or use her keys. She says this problem began upon awakening. On exam, there were no abnormalities of strength, sensation, or coordination. Which of the following is the term for this disorder?

- a. Malingering
- b. Prosopagnosia
- c. Apraxia
- d. Astereognosia

65. Which of the following features would best distinguish inflammatory muscle diseases from muscular dystrophies?

- a. Patients with inflammatory muscle disease may have an associated skin rash unlike those with muscular dystrophies who have no dermatological features
- b. In inflammatory muscle disease there is usually a family history of the disease which is not the case in muscular dystrophies
- c. Muscle wasting is more marked in inflammatory muscle disease than in muscular dystrophies
- d. Muscular dystrophies progress quite rapidly unlike inflammatory muscle disease is slow

66. During motor system examination, which of the following best describes the kind of increased tone called rigidity:

- a. Cog wheel
- b. Rubber pulley
- c. Lead pipe
- d. Clasp knife

67. In Parkinson's disease which is the typical gait disorder?

- a. Waddling gait

- b. Apraxic gait
- c. Festinating gait
- d. Spastic gait

68. Which of the following diseases might account for a patient presenting with weakness, hypotonia (flaccidity), hyporeflexia, atrophy and fasciculations in her right lower limb?

- a. Parasagittal meningioma
- b. Transverse myelitis
- c. Stroke
- d. Chronic inflammatory demyelinating polyneuropathy

69. Regarding the presentation of neurological diseases which of the following is TRUE?

- a. Rapid progression over days is a feature of degenerative disease
- b. Recurrent discrete episodes of symptoms over months to years suggests an infection
- c. Remitting and relapsing symptoms implies a space-occupying lesion (SOL)
- d. A sudden presentation with rapid recovery occurring suggests a transient ischaemic attack

70. Which of the following secondary symptoms is most frequently caused by idiopathic intracranial hypertension?

- a. Conductive hearing loss
- b. Pulsatile tinnitus
- c. Sensorineural hearing loss
- d. Palatal myoclonus

71. Which of the following best characterizes temporal lobe epilepsy?

- a. Complex partial seizures characterized by aura
- b. Sudden lapses in muscle tone resulting in lapses in posture or falls
- c. An abrupt onset loss of consciousness, a vacant look and behavioural arrest

d. Provocation of seizures by pouring hot water over head during bathing

72. Which of the following patients is most likely suffering from headaches due to temporal arteritis?

a. 25-year-old female. Pain is bilateral, constant, no nausea, no neurological deficits. Relieved by NSAIDs.

b. 75-year-old male. Pain throbbing and over frontotemporal region, history of polymyalgia rheumatica.

c. 40-year-old male. Pain is sharp, and occipital in region, associated with vertigo and ataxia.

d. 45-year-old female. Pain is sudden onset and global, had a loss of consciousness at the time, vomiting, no neurological deficit.

73. Regarding Myasthenia Gravis which of the follow is NOT TRUE?

a. Poor peak flow, weak cough and hypercapnoea are reasons for ICU or intubation

b. Depolarizing muscle relaxants have a prolonged duration of action when used in Myasthenia Gravis but this is not the case with non-depolarizing muscle relaxants

c. Giving edrophonium for diagnosis can cause cholinergic crisis if the cause of weakness is not Myasthenia Gravis

d. The presence of anti-MuSK antibodies is poorly predictive of the diagnosis

74. In which of these patients is the diagnosis of Idiopathic intracranial hypertension most likely correct?

a. 60-year-old male. Pain is diffuse, rapid onset, associated with neck stiffness, photophobia. Afebrile in ED but did describe shivering at home.

b. 65-year-old male. Pain dull over forehead, worsened when leaning forward, recent URTI symptoms, fever at 38 degrees.

c. 35-year-old female. Pain is diffuse, gradual onset, associated with vomits and started few days after birth of 1st baby

d. 30-year-old female. Pain has been ongoing for weeks, constant and worsened by coughing, associated nausea and blurred vision, BMI 34

75. A 26-year-old male patient presents with a power of 3 in his lower limbs, depressed reflexes and muscle wasting. Which of the following features would put the site of the lesion at the level of the muscle?

- a. Predominantly proximal muscle weakness
- b. Abnormal nerve conduction test results
- c. The presence of fasciculation
- d. Muscle percussion-induced contraction

76. Which is TRUE about status epilepticus?

- a. Status is defined as a single seizure lasting ≥ 10 minutes
- b. Ketamine is contraindicated in seizures due to risk of raised intracranial pressure
- c. Correctable metabolic causes of seizure include hyponatraemia, hypoglycaemia and hypocalcaemia
- d. ≥ 2 seizures within one ED presentation is considered status epilepticus regardless of post ictal recovery in between

77. Difficulty identifying tactile sensations of written patterns (shapes or letters) on the skin is known as ...

- a. Simultagnosia
- b. Stereognosis
- c. Graphaesthesia
- d. Praxis

78. An 80-year-old man sees you for slurred speech, which is evident on office examination. Notable findings include deviation of his tongue to the left, when asked to stick his tongue out straight. The left side of the tongue looks withered and “twitchy.” The rest of his neurological examination is normal. Where is the most likely lesion here?

- a. left glossopharyngeal nerve
- b. left hypoglossal nerve

- c. left facial nerve
- d. right frontal motor cortex

79. A patient is asked to demonstrate how to use a hammer. Instead of demonstrating the use of a hammer, the patient simulates cutting with a saw. Which of the following correctly identifies this condition?

- a. Graphasthesia
- b. Apraxia
- c. Astereognosis
- d. Mutism

80. A 33 year-old severely obese patient with a long history of severe intermittent headaches unresponsive to several routine analgesics feels much better after undergoing a lumbar puncture as part of her neurology work up. Which of the following is her most likely diagnosis?

- a. Venous sinus thrombosis
- b. Cluster Headache
- c. Benign Intracranial Hypertension
- d. Migraine

81. A 56-year-old woman with a history of alcohol abuse presented with an unsteady gait for the past year. On exam, she had a wide-based gait and fell if stood with her feet together and eyes closed. She was mildly ataxic on finger–nose–finger testing and her strength was intact. Which of the following is most likely to be seen on a brain MRI?

- a. Atrophy of the cerebellar vermis
- b. Midbrain atrophy
- c. Degeneration of the inferior olives in the medulla
- d. Hydrocephalus and narrowing of the cerebral aqueduct

SURGERY

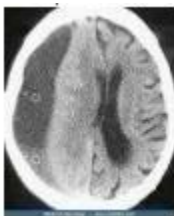
82. A 30-year-old male road accident victim had a non-contrasted brain CT scan done which showed an intracranial bleed which was described as being a biconvex lesion, which is hyperintense to brain. What condition does the patient have?

- a. Acute subdural haematoma
- b. Chronic extradural haematoma
- c. Chronic subdural haematoma
- d. Acute extradural haematoma

83. A 20 year old patient who was previously stabbed in his head is now having intermittent flashes of light which was confirmed by the Neurologist as a form of seizure. On which part of the head was the patient most likely stabbed?

- a. Face
- b. Back of the head
- c. Through the frontal air sinus
- d. Side of the head above the ear

84. A 70 year old female patient knocked her head against a wall 6 weeks ago. She has over the past one (1) week progressively developed a hemiparesis and reduced level of consciousness. What condition is depicted?



- a. Chronic subdural haematoma
- b. Acute subdural empyema
- c. Acute subdural haematoma
- d. Chronic subdural empyema

85. A diagnostic image of a brain injured patient was described by the radiologist as a parenchymal round rim enhancing lesion after contrast with a dark core: What condition does the patient have?

- a. Brain contusion
- b. Cerebral hematoma
- c. Brain abscess
- d. Cerebral cyst

86. Which of the following conditions is least likely to be a long-term complication of concussion:

- a. Chest pain
- b. Sleeplessness
- c. Loss of libido
- d. Chronic headache

87. The best predictor of outcome in a severely brain injured patient is the:

- a. Overall GCS
- b. Best motor response
- c. Verbal response
- d. Eye opening response

88. The following factor is a possible cause of adult acquired obstructive hydrocephalus:

- a. Cytomegalovirus infection in pregnancy
- b. Subarachnoid hemorrhage
- c. Dandy-walker malformation
- d. Tuberculous meningitis

89. A patient who is assessed to be able to correctly localize pain, opens eyes to pain and moans to pain and calls is said to have a GCS of:

- a. 9/15
- b. 10/15
- c. 7/15
- d. 8/15

90. The following history or physical examination finding in a road accident victim is a strong indication for a brain CT scan:

- a. BP 70/40mmHg
- b. Patient was an unhelmeted motor biker
- c. GCS 15/15
- d. Has Arterial Oxygen Saturation of 70%

91. An overweight 40 year old lady presents with a history of intermittent headaches for the last six months is found to have papilloedema on clinical examination. Which of the following is the LEAST likely diagnosis?

- a. Malignant hypertension
- b. Benign intercranial hypertension
- c. Tension Headache
- d. Tumor

92. A 4 year old patient known with congenital hydrocephalus and a VP shunt in situ now presents with a 3 day history of headaches followed by reduced level of consciousness, no fever. What is the most likely cause of his condition?

- a. Calcification of the ventricular catheter
- b. VP shunt blockage
- c. Intestinal obstruction from adhesions
- d. VP shunt infection

93. The following maneuver can be done to test gag reflex in a patient who is intubated and being ventilated in the Intensive Care Unit:

- a. Depressing the tongue and touching the throat with a tough depressor
- b. Depressing the tongue and touching the throat with a wound swab
- c. Yanking on the securely halted E-tube
- d. Blowing warm air into the E-tube

94. A 15 year old patient who fell from a coconut tree 2 years ago now presents to the OPD with a one week history of tonic-clonic seizures: Initial investigations to undertake include:

- a. Brain CT scan
- b. Brain tractography
- c. Electrocardrogram
- d. Echocardiogram

95. The following activity is not a pre-requisite when performing brain death certification:

- a. Confirming that the patient definitely has a traumatic brain injury
- b. Ensuring that the patient is not hypothermic
- c. Ensuring that the patient has not taken any severely sedating drugs.
- d. Confirming the GCS to 3/15

96. What is the most common cause of a traumatic lumbar puncture?

- a. Bacterial infection
- b. Viral infection
- c. Contaminated needle
- d. Red blood cell contamination

97. During a focused assessment of a patient with suspected brain injury, the following is LEAST important.

- a. Taking a focused history
- b. Testing all the cranial nerves
- c. Checking the Glasgow coma score
- d. Checking the pupil sizes

98. Which of the following patients with diffuse brain injury is deemed to have an absolute indication for intubation and ventilation? The Patient with GCS:

- a. 1/5-8/15
- b. 3/15-14/15
- c. 3/15-8/15
- d. 3/15-10/15

99. The following manoeuvre is NOT a possible measure to prevent hypoxia in a brain injured patient:

- a. Removal of all foreign bodies blocking their airway
- b. Removal of patients clothing and neck collar
- c. Intubation and ventilation
- d. Administering oxygen through a non-rebreather mask

100. A 32 year female patient who fell from a height 3 years ago is now reporting to the OPD with two episodes of focal left upper limb seizures over the past 4 days. Which of the following findings on imaging could point to the underlying cause?

- a. Temporal lobe spongiosis
- b. Occipital lobe scar
- c. Frontal lobe spongiosis
- d. Parietal lobe spongiosis

101. Regarding the subarachnoid cisterns:

- a. The cisterna magna lies between the pons and the midbrain
- b. The vertebral artery lies in the pontine cistern.
- c. The cisterna magna is continuous below with the spinal subarachnoid space.
- d. The cisterna magna contains the pons and posterior inferior cerebellar arteries

102. A 25 year old male patient with traumatic brain injury was assessed to have a GCS score 6/15 His neurological status is as follows:

- a. Localises to pain opens eyes to calls and moans
- b. Abnormally flexes to pain opens eyes spontaneously no verbal response
- c. Localises pain opens eyes to calls and confused
- d. Abnormally flexes to pain opens eyes to pain no verbal response

103. Which of the following patients is MOST likely to have the best overall outcome after being diagnosed with severe diffuse brain injury:

- a. GCS7/15 - M-5, V-1, E-1
- b. GCS7/15 - M-2, V-2, E-3
- c. GCS6/15 - M-4, V-1, E-1
- d. GCS7/15 - M-2, V-1, E-4

104. An elderly woman was brought to the emergency room with a history of fall. She is normotensive. Her GCS was 11, no focal neurological deficit. If the CT scan showed hyperdensities in the suprasellar and prepontine cisterns, these features will be consistent with:

- a. Acute subarachnoid hemorrhage
- b. Acute epidural hematoma
- c. Chronic subdural hematoma
- d. Acute subdural hematoma

105. The following imaging feature will NOT be suggestive of generalized brain swelling after a brain injury:

- a. Attenuated basal cisterns
- b. Absent basal cisterns
- c. Obliterated sulci-gyri markings
- d. Obstructed foramina of Monroe

106. A patient who is assessed to abnormally flex to pain on the right and abnormally extend to pain on the left is said to have a motor score of:

- a. 1
- b. 3
- c. 4
- d. 2

107. A 30-year-old male footballer who crashed on the field of play with another player had a 30 seconds loss of consciousness and one minute duration of anterograde memory loss. What condition did he have?

- a. Severe diffuse brain injury
- b. Consolidation
- c. Mild focal brain injury
- d. Concussion

108. A 30-year-old patient with confirmed diffuse brain injury has now been found to be passing large volumes of urine with high sodium concentration and high osmolarity. The most likely cause of his current condition is:

- a. Syndrome of inappropriate ADH production
- b. Diabetes mellitus
- c. Diabetes insipidus
- d. Syndrome X

109. The following condition is a cause of communicating hydrocephalus:

- a. Subarachnoid haemorrhage
- b. Pineal gland tumour
- c. Intraventricular haematoma
- d. Aqueduct stenosis

110. The following is a practical way to increase one's suspicion index of a CSF leak in a traumatic brain injury patient with suspected base of skull fracture:

- a. blood soaking patients' pillow case
- b. Bull's eye sign of fluid on patients' pillow case
- c. fever on day one of arrival
- d. profuse bleeding from patient's ear

111. The following measure can be instituted to reduce intracranial hypertension in a brain injured patient:

- a. Raising the foot of the bed 30 degrees
- b. Turning the neck to the side of the brain injury
- c. Raising the head of the bed 30 degrees
- d. Turning the neck away from the side of the brain injury

112. A 25-year-old patient with a confirmed diffuse brain injury from a fall from height is spiking temperatures. The following measure can be used to BEST control the fever:

- a. Giving paracetamol and doing tepid sponging
- b. Fanning the patient only
- c. Putting on the fan and giving cold IV fluids
- d. Fanning the patient plus giving oral fluids

113. A patient with very high fevers just after a severe post traumatic diffuse brain injury is most likely caused by:

- a. Dysregulation of the body's temperature
- b. Dysregulation of the patient's entire nervous system
- c. Dysregulation of the patients' autonomic nervous system
- d. Total shutdown of the patients' immune system and subsequent infection

114. The following cranial nerve palsy is most likely to occur in diffuse brain injury with raised intracranial pressure. Cranial nerve:

- a. II
- b. I
- c. IV
- d. VII

115. The following eye sign is a possible pointer to an acutely raised intracranial pressure in any patient.

- a. Optic disc pallor
- b. Optic disc atrophy
- c. Periorbitaloedema
- d. Papilloedema

116. A 32year old man fell from a scaffolding at a construction site hitting his head. At the emergency room, he has lucid intervals prior to rapid deterioration. A non-contrast CT scan of the brain shows an extra-axial, well-defined ellipsoid area of hyperdensity and causing midline shift to the left. These findings are consistent with:

- a. Acute Subarachnoid hematoma
- b. Acute right frontal lobe hematoma
- c. Acute epidural hematoma
- d. Acute subdural hematoma

117. A 25 year old unhelmeted motor biker crashed into a stationary vehicle and conveyed to the accident and emergency unit. In your initial assessment, the following information will NOT be enquired about.

- a. When did the patient eat last and what exactly did he eat?
- b. What happened?
- c. When did the accident happen?
- d. What has the patient's condition been from the time of the incident till now?

118. The following eye sign is the best pointer to a chronically raised intracranial pressure in a patient with a known chronic brain pathology:

- a. Papilloedema
- b. Peri orbital atrophy
- c. Optic disc atrophy
- d. Peri orbital oedema

119. The following history or physical examination finding on a patient in the accident and emergency center is the weakest indication for a brain CT scan:

- a. A known intoxicated alcoholic found unconscious by the road side
- b. Glasgow coma score of 9/15 - ; V-2; E-2
- c. brushed head against the wall with mild headache
- d. Knocked down by a bus with some headache

120. A 15 year old female patient knocked down by a motor bike has base of skull fracture after imaging. Which of the following signs is LEAST likely to be found in her?

- a. Optic nerve atrophy
- b. Racoon eyes
- c. Facial nerve palsy
- d. Battle sign

121. The following condition can be a direct complication of open skull fracture of the cranial vault:

- a. CSF otorrhea
- b. Retinal detachment
- c. Cerebritis
- d. CSF rhinorrhea

122. In the event of a patient having a confirmed space occupying supratentorial lesion with a unilateral fixed dilated pupil, the following system is compromised:

- a. Sympathetic nervous system on cranial nerve III
- b. Parasympathetic nervous system on cranial III
- c. Autonomic nervous system on cranial nerve III
- d. Parasympathetic nervous system on cranial nerve

123. The following factor is NOT a cause of secondary brain injury in a road traffic accident victim.

- a. Prolonged uncontrolled seizures
- b. Persistently low arterial oxygen saturation
- c. Hitting of patient's head against the steering wheel
- d. Persistently low blood pressures

124. A 67-year-old man was involved in high impact accident. Complains of neck pains on arrival at the emergency room. A CT scan of the cervical spine shows a fracture through the base of the dens. This is classified as:

- a. Odontoid fracture Type I
- b. Odontoid fracture Type IV
- c. Odontoid fracture Type II
- d. Odontoid fracture Type III

125. A young man dived head first into a shallow swimming pool. At the accident and emergency department, he complains of inability to move his lower limbs. The patient inability to move the limbs is suspicious for spinal cord and ligamentous injury. What is the best imaging modality to confirm such a diagnosis?

- a. Lateral and open mouth radiographs of the cervical spine
- b. Contrast CT scan of the cervical spine
- c. Non-contrast CT scan of the cervical spine
- d. MRI of the cervical spine

126. The most important measure to put in place to prevent re-bleeding in the management of a patient with haemorrhagic stroke is:

- a. Control of high temperatures
- b. Intubation and Ventilation
- c. Control of high blood pressures
- d. Airway Management

127. A 48-year-old lady is transported to the emergency room after awakening with a severe "thunderclap" headache and stiff neck. There are no signs of hemorrhage in the brain on head computed tomography (CT) scan. Which of the following is the ideal next move?

- a. Lumbar puncture
- b. Ergotamine therapy
- c. Electroencephalography (EEG)
- d. Venography

128. A 40 years old patient with a contre coup injury was found to be bleeding profusely from a scalp laceration. Where on the scalp is he most likely to be bleeding from?

- a. Parietal area
- b. Occipital area

- c. Forehead
- d. Temporal area

129. A 22-year-old student Nurse lifted her fridge and immediately lost power in both legs with loss of bowel and bladder control. Which condition is best described by her clinical features?

- a. Cervical myelopathy
- b. Spine fractures
- c. Cauda equine syndrome
- d. Sudden spondylosis syndrome

130. The following treatment measure is neuroprotective in the event of a brain injury:

- a. Therapeutic hypothermia
- b. Extended use of mannitol
- c. Prolonged use of steroids
- d. Therapeutic hyperthermia

131. Which of the following factors is most likely to cause an increase in cerebral blood flow and increase in ICP in a patient with severe diffuse brain injury? Arterial:

- a. High O₂
- b. High Na⁺
- c. High CO₂
- d. Low CO₂

132. A patient with severe diffuse brain injury was admitted to Accident and Emergency Unit yesterday. What feeding route will be the best for this patient?

- a. Dextrose IV fluids
- b. Feeding per NG tube
- c. Total parenteral feeding

d. No feeding is beneficial

133. A 12-year-old patient with fever, neck stiffness and a focal right arm and leg power deficit has presented to the accident and emergency unit. The following immediate assessment is NOT indicated.

- a. Estimation of Glasgow Coma score
- b. Brain CT scan without and with contrast
- c. Brain CT scan with contrast
- d. Lumbar puncture for cerebrospinal fluid

134. A nurse at female ward has just called the doctor on duty to see a patient initially admitted with a stroke who has just dropped her GCS from 15/15 to 7/15. Initial management to institute include:

- a. Application of face mask and administering oxygen
- b. Use of oxygen non-rebreather mask
- c. Giving of oxygen through nasal prongs
- d. Intubation and ventilation

PATHOLOGY

135. The following imaging feature is highly suggestive of malignancy of a brain tumour:

- a. Definability of the edges
- b. Cystic component
- c. Large size
- d. Necrosis

136. A 54 year old farmer is brought by his wife into the outpatient clinic with complaints that he has become very sluggish. On examination the attending doctor finds increased tone in the limbs and notices tremors in both hands. What pathological changes would you expect in this patient?

- a. Cerebral inclusion bodies called Negri bodies
- b. Amyloid plaques and neurofibrillary tangles
- c. Demyelination in extrapontine regions
- d. Loss of neurons in the substantia nigra pars compacta

137. Diffuse plaques, are defined by all the following EXCEPT:

- a. dominated by amyloid beta type 40
- b. represent an early stage of plaque development
- c. Absence of surrounding neuritic reaction
- d. are found in superficial portions of cerebral cortex as well as in basal ganglia and cerebellar cortex.

138. Though the diagnostic features of Alzheimer's disease (AD) relate to the presence of plaques and tangles, other pathologic findings are seen in the setting of AD include all the following EXCEPT:

- a. Granulovacuolar degeneration
- b. Accumulation of tau proteins.
- c. Hirano bodies
- d. Cerebral amyloid angiopathy

139. In relation to Alzheimer's plaques, other proteins that are present in plaques in lesser abundance, include all the following EXCEPT:

- a. apolipoproteins
- b. α_1 -antichymotrypsin
- c. Amyloid Beta
- d. components of the complement cascade

140. In relation to Alzheimer's disease, plaques are usually less prominent in the:

- a. Hippocampus
- b. Amygdala
- c. Neocortex
- d. primary motor cortices

141. In the involvement of the brain in Alzheimer's Disease (AD), which of the following is the CORRECT sequence of events?

- a. ADs evident earliest in the hippocampal formation and isocortex, and then extend into the neocortex, finally involving the entorhinal cortex
- b. AD is evident earliest in the the neocortex then the entorhinal cortex, then spread through the hippocampal formation and isocortex
- c. AD is evident earliest in the isocortex then the entorhinal cortex, then spread through the hippocampal formation and then extend into the neocortex
- d. AD is evident earliest in the entorhinal cortex, then spread through the hippocampal formation and isocortex, and then extend into the neocortex

142. Multiple sclerosis is a disease that results in the formation of plaques within the brain. These plaques are most prominently located in the:

- a. In the ventricles
- b. White matter
- c. Gray matter
- d. At the gray matter white matter junction

143. In relation to Alzheimer's disease, neurofibrillary tangles are usually less prominent in the:

- a. Amygdala
- b. Neocortex
- c. Primary motor cortices
- d. Hippocampus

144. A mother complains to you that her 8yr old boy has started walking like a rogue and will not stop even when cautioned... Your examination further reveals the child has balance and gait abnormalities and may actually be having signs of increased intracranial pressure from an intracranial tumour. To confirm this diagnosis, the pathologist will request for:

- a. PAS special stain
- b. Immunohistochemical staining with GFAP
- c. Giemsa stain
- d. ZN stain for acid fast bacilli

145. Assuming that histopathologically, this lesion is described as possessing areas of necrosis with pseudopalisading and Glomeruloid bodies, it means that it is:

- a. An anaplastic astrocytoma
- b. Pilocytic astrocytoma
- c. Glioblastoma
- d. A pleomorphic xanthoastrocytoma

146. A pilocytic astrocytoma is classified according to the WHO grading system as:

- a. A grade III/IV tumour
- b. A grade I/IV tumour
- c. A grade IV/IV tumour
- d. A grade II/IV tumour

147. This grading supposes that the tumour has:

- a. An Intermediate prognosis
- b. A good prognosis
- c. None of the other options
- d. A poor prognosis

148. A 4th year medical student wishes to know the likely cell in the brain that may be involved. From your examination and history, you will explain to her that the cells of interest are:

- a. Astrocytes
- b. Ependymal cells
- c. Meningothelial cells
- d. Oligodendrocytes

149. Supposing this child had an intracranial space occupying lesion, it is most likely to be:

- a. A glioma
- b. An abscess
- c. A meningioma
- d. A tuberculoma

150. You suspect that this child has increased intracranial pressure, your biggest fear is that the brain will herniate. This is most likely to be a:

- a. Tonsillar herniation
- b. Subfalcine herniation
- c. Mesial temporal herniation
- d. Any of the other options

151. With foci of necrosis, pseudopalisading and glomeruloid bodies, this lesion will be graded according to WHO grading as:

- a. grade I/IV tumour
- b. A grade II/IV tumour
- c. A grade III/IV tumour
- d. A grade IV/IV tumour

152. In this form of herniation the brain herniates:

- a. Through the foramen magnum
- b. Under the tentorium cerebelli
- c. Under the falx cerebri
- d. None of the other options

153. A glioma in this age group is likely to be an:

- a. Undifferentiated tumour
- b. Oligodendrogliomas
- c. Ependymoma
- d. Pilocytic astrocytoma

154. Another 4th year student is fascinated by the word 'glioma' and wishes to know what it means. You will explain that these tumours arise from:

- a. Meningothelial cells, ependymal cells, astrocytes and oligodendrocytes
- b. Macroglial cells and microglial cells
- c. Ependymal cells, astrocytes and microglial cells
- d. Ependymal cells, astrocytes and oligodendrocytes

155. After a bat bite, a rural farmer suffers from flu-like symptoms. Over the following days he develops diffuse weakness and confusion and is dead within 1 month. Histopathology was done on brain specimens taken from autopsy. What is the name of the pathognomonic findings likely found in this patient?

- a. Howell-Jolly bodies
- b. Negri bodies
- c. Rosenthal fiber
- d. Cowdry type A

156. The neurosurgeon enucleates the lesion and sends it for histopathological analysis. The pathologist describes a lesion that has chronic granulomatous inflammation with caseous necrosis. His diagnosis will likely be:

- a. A meningioma
- b. An abscess
- c. A tuberculoma
- d. A glioma

157. You are convinced that this is a pilocytic astrocytoma. You expect the pathologist to tell you the lesion is a tumour with:

- a. Papillary elements in a myxoid background admixed with cells with regular, round to oval nuclei with abundant granular chromatin
- b. Bipolar cells with long, thin hairlike processes that are GFAP positive; Rosenthal fibres, eosinophilic granular bodies and microcysts
- c. Areas of necrosis and haemorrhage with pseudopalisading around these necrotic areas
- d. Sheets of regular cells with spherical nuclei containing finely granular chromatin surrounded by a clear halo of cytoplasm

158. Supposing this is a glioma in this age group it is most likely to be:

- a. Spinal
- b. Infra-tentorial
- c. Supra-tentorial
- d. Subdural

159. What is the significance of the APOE $\epsilon 4$ allele?

- a. It is a structural protein found in neurons.
- b. It increases the risk of Alzheimer's disease.
- c. It stabilizes neuronal axons.
- d. It forms plaques in the brains of Alzheimer's patients.

160. What is the role of Tau protein in neurons?

- a. It forms plaques in Alzheimer's disease.
- b. It is involved in lipid metabolism.
- c. It stabilizes neuronal axons.
- d. It is released into the bloodstream upon neuronal injury.

161. Which protein forms plaques in the brains of individuals with Alzheimer's disease?

- a. Neurofilament Light Chains (NfL)
- b. Amyloid Beta ($A\beta$)
- c. APOE $\epsilon 4$
- d. Tau protein

162. Which of the following is true about Neurofilament Light Chains (NfL)?

- a. They stabilize neuronal axons.
- b. They are structural proteins found in neurons.
- c. They are only found in the bloodstream.
- d. Elevated levels are specific to Alzheimer's disease.

163. Which of the ff astrocytic tumours has the worse prognosis?

- a. diffuse astrocytoma
- b. glioblastoma
- c. pilocytic astrocytoma
- d. pilomyxoid astrocytoma

164. Mr Mensah a chronic alcohol died suddenly in his room after a bout of alcoholic intoxication. At autopsy there is a massive subdural haemorrhage. The pathologist on duty tried dating the hematoma. Which of the ff is a feature of a two week old hematoma?

- a. calcification of hematoma
- b. early dev of hyalinized connective tissue
- c. growth of fibroblast from the dural surface
- d. lysis of the clot

165. a 45yr old man with brain tumour underwent ta sirgeru and sample sent for histopathology. The diagnosis came as ODG (oligodendroglioma). which of the ff is a feature of this tumour?

- a. perinecrotic palisading of tumour cells
- b. the tumour cells are spindle with bipolar hair-like process
- c. the tumour cells have central n. with a halo clear cytoplasm
- d. whorl or sheets of meningotheial cells

PEDIATRICS

166. An infant is noted to have infantile spasms. EEG classically shows a hypsarrhythmia. The pediatric house officer noted cutaneous lesions at the back which she noted as ash-leaf depigmentation. The disorder MOST LIKELY to affect this infant is:

- a. Psoriasis
- b. Neurofibromatosis
- c. Tuberous sclerosis
- d. Incontinentiapiementi

167. A 4 year old patient known with congenital hydrocephalus and a VP shunt in situ now presents with a 3 day history of headaches followed by reduced level of consciousness, no fever. What is the most likely cause of his condition?

- a. Calcification of the ventricular catheter
- b. VP shunt blockage
- c. Intestinal obstruction from adhesions

d. VP shunt infection

168. A 9 year old child known with a myelomeningocele otherwise well now presents with loss of bowel and bladder function and paraparesis of progressive nature. What is the most likely cause of his condition:

- a. Tethering of the spinal cord
- b. New spinal cord herniation
- c. Necrosis of the distal spinal cord
- d. Spondylosis of lumbo-sacral spine

169. Regarding childhood absence epilepsy, which of the following is FALSE:

- a. Developmentally normal
- b. EEG, symmetrical bilateral spike/wave discharges at a frequency of 6Hz
- c. Onset 3-12yrs
- d. Frequent absence seizures lasting 5-20 sec

170. A 7-year old awakens at 02:00 screaming. His parents report that he was hyperventilating at the time and had dilated pupils. He was inconsolable and unaware of the environment. After a few minutes, he returns to sleep and recalls nothing the following morning. Which of the following is the most likely diagnosis?

- a. Drug ingestion
- b. Night terrors
- c. Seizure disorder
- d. Munchausen syndrome by proxy

171. A 19 day old infant presents with fever, irritability, poor feeding and a bulging fontanelle. CSF evaluation demonstrates gram positive cocci. The most likely causative agent is:

- a. *Listeria monocytogenes*
- b. *Haemophilus influenzae*

- c. Group B streptococcus
- d. Group A streptococcus

172. A 3-month-old baby with a rapidly enlarging head had a head scan which showed enlargement of the lateral ventricles and the third ventricle but a small fourth ventricle. What condition does the baby most likely have?

- a. Bifid septum pellucidum
- b. Stenosis of foramen Monroe
- c. Dandy-walker malformation
- d. Stenosis of Aqueduct of Sylvius

173. The following patient position is most highly recommended for nursing a baby with a leaking myelomeningocele:

- a. Recovery
- b. Lithotomy
- c. Prone
- d. Supine

174. Choose the TRUE statement concerning Cerebral Palsy in children.

- a. The Cerebellar region of the brain is the most affected
- b. Bulbar palsy is common
- c. Spastic diplegia is less common among preterms
- d. Hand preference before 12months of age can be a useful sign

175. The Floppy infant is LEAST likely to have which of the following as a cause?

- a. Prader Willi Syndrome
- b. Turners Syndrome
- c. Trisomy 21

d. Spinal muscular atrophy

176. Which of the following is FALSE about febrile convulsions?

- a. By definition occurs in children between 6 months and 5 years.
- b. Rectal diazepam is usually required to stop seizures
- c. Recurrence rate is about 3%
- d. A urinalysis is important

177. A 2 year old child with tb meningitis now has reduced level of consciousness and bilateral cranial nerve VI palsies. Ct brain done showed enlarged lateral and third ventricles but a small fourth ventricle. The diagnosis is:

- a. Acquired communicating hydrocephalus
- b. Acquired obstructive hydrocephalus
- c. Chiari malformation
- d. Macrocephaly

178. A 5 month old baby was brought to the Paediatric clinic with a rapidly increasing head circumference, splayed sutures, bulging anterior fontanelle and a sun setting sign. CT brain scan showed enlarged lateral and third ventricles but a small fourth ventricle. This clinical and radiological picture is that of:

- a. Congenital communicating hydrocephalus
- b. Acquired communicating hydrocephalus
- c. Congenital obstructive hydrocephalus
- d. Acquired obstructive hydrocephalus

179. The following factors predisposes a newly pregnant woman to having a child with myelomeningocele EXCEPT:

- a. Maternal diabetes
- b. Maternal fever in pregnancy

- c. Previous child with myelomeningocele
- d. Maternal hypertension

PHARMACOLOGY

180. The addition of a vasoconstrictor to prolong duration of action would not be relevant when which of these local anaesthetics is/are being used:

- I. Cocaine
 - II. Ropivacaine
 - III. Lidocaine
- a. I and III
 - b. I
 - c. I and II
 - d. III

181. Originally developed as an antiviral drug, is now used as adjunct therapy for dyskinesia produced by levodopa:

- a. Entacapone
- b. Amantadine
- c. Tolcapone
- d. Carbidopa

182. Auto-induction of metabolism is particularly troublesome with this anticonvulsant and requires dosage adjustment with prolonged usage:

- a. Valproate
- b. Ethosuximide
- c. Carbamazepine
- d. Phenytoin

183. Which of the following drugs is NOT a selective serotonin reuptake inhibitor?

- a. Fluvoxamine
- b. Sertraline
- c. Fluphenazine
- d. Fluoxetine

184. Gingival hyperplasia and hirsutism are characteristic adverse effects of:

- a. Levetiracetam
- b. Phenytoin
- c. Valproic acid
- d. Carbamazepine

185. Though a very potent anaesthetic, the use of this agent is limited by extensive nephrotoxicity:

- a. Nitrous oxide
- b. Methoxyflurane
- c. Isoflurane
- d. Ketamine

186. Which of the following statements is UNTRUE about glutamate receptors in the CNS:

- a. K^+ blocks the NMDA type glutamate receptors.
- b. Metabotropic glutamate receptors can mediate both inhibitory and excitatory effects
- c. AMPA receptor subtype has a co-modulatory binding site for glycine
- d. NMDA receptor subtype gates both Na^+ and Ca^{2+}

187. Which of the following anti-parkinson drugs is a DOPA decarboxylase inhibitor?

- a. Selegiline
- b. Levodopa
- c. Benserazide
- d. Bromocriptine

188. Which of the following statements about memantine therapy is INCORRECT?

- a. It allows patients to maintain certain daily functions a little longer than they would without the medication.
- b. It is used to treat moderate to severe Alzheimer's disease.
- c. It may delay the worsening of symptoms in some people.
- d. It regulates the activity of glutamate in the brain by stimulating NMDA receptors.

189. Pethidine is preferred over morphine in obstetrics because pethidine:

- I. Does not require glucuronide conjugation prior to elimination
 - II. Has a short duration of action
 - III. Has antimuscarinic effects
- a. III
 - b. I
 - c. I and III
 - d. I and II

190. The neurotransmitter, 5-HT, is significantly involved in all the following neurological responses EXCEPT:

- a. feeding behavior.
- b. respiration.
- c. control of mood.
- d. vomiting.

191. Antidepressant can also be used for the following indications EXCEPT:

- a. Chronic pain
- b. Attention deficit hyperkinetic disorder
- c. Diarrhoea
- d. Enuresis

192. The following are cholinesterase inhibitors most commonly prescribed for mild to moderate Alzheimer's disease EXCEPT:

- a. Galantamine
- b. Donepezil
- c. Rivastigmine
- d. Neostigmine

193. Behavioral and psychological symptoms of dementia in people with Alzheimer's can be managed with all of the following EXCEPT:

- a. Mood stabilizers e.g. carbamazepine
- b. MAO inhibitors e.g. tranylcypromine
- c. Antipsychotics e.g. haloperidol
- d. Antidepressants e.g. fluoxetine

194. All the following agents can be used safely in the treatment of absence seizures EXCEPT:

- a. Lamotrigine
- b. Valproic acid
- c. Carbamazepine
- d. Ethosuximide

195. Which of the following is the appropriate molecular target for the AED useful in super-refractory status epilepticus, lacosamide:

- a. GABA transaminase
- b. SV2A
- c. CRMP-2
- d. AMPA receptors

196. Which of the following statements about the GABAA receptor is NOT true?

- a. It is the molecular target for the drug flumazenil
- b. The receptor gates Cl⁻ influx
- c. Benzodiazepines and barbiturates bind to the same site on the receptor
- d. It mediates fast inhibitory synaptic transmission in the CNS

197. Which of the following medications acts on N-methyl-D-aspartate (NMDA) glutamate receptors?

- a. Memantine
- b. Tacrine
- c. Rivastigmine
- d. Galantamine

198. The atypical antidepressant drug, trazadone, blocks all the following receptors EXCEPT:

- a. H1
- b. 5-HT_{2A}
- c. D2
- d. 5-HT_{2C}

199. Which of the following anticonvulsants is INAPPROPRIATELY matched with its mechanism of action?

- a. Carbamazepam — inhibition of T-type Ca²⁺ channels
- b. Levitiracetam - inhibition of synaptic protein SV2A

- c. Vigabatrin — inhibition of GABA transaminase
- d. Tiagabine - inhibition of GABA transporter (GAT1)

200. All the following anticonvulsants may exacerbate absence seizure except:

- a. Carbamazepine
- b. Phenytoin
- c. Valproate
- d. Felbamate

201. Which of the following opioid does not produce the characteristic pinpoint pupils observed in the use of opioids?

- a. Methadone
- b. Pethidine
- c. Heroin
- d. Codeine

202. Which of the following would likely lead to an infusion syndrome characterized by rhabdomyolysis, metabolic acidosis and cardiovascular collapse?

- a. Fentanyl
- b. Propofol
- c. Ketamine
- d. Thiopentone

203. Stimulating 5-HT auto-receptors will result in:

- a. generation of excitatory post synaptic potentials.
- b. inhibition of 5-HT storage in synaptic vesicles.
- c. reduced breakdown of 5-HT at the synaptic cleft
- d. reduced 5-HT release into synapse.

204. The use of this agent as an anticonvulsant is limited to the management of only catamenial epilepsy?

- a. Gabapentin
- b. Acetazolamide
- c. Magnesium sulphate
- d. Topiramate

205. Which of the following drugs is not an irreversible non-selective monoamine oxidase inhibitor?

- a. Tranylcypromine
- b. Isocarboxazid
- c. Moclobemide
- d. Phenelzine

206. A boy, age 8, frequently displays bouts of gazing while in class. What drug is best for this patient out of the ones listed below?

- a. Phenytoin
- b. Levetiracetam
- c. Valproic acid
- d. Ethosuximide

207. Which of the following is NOT TRUE about amantadine?

- a. It enhances reuptake of released dopamine by pre-synaptic neuron
- b. It is well absorbed following oral administration and excreted unchanged by the kidney
- c. It promotes the release of dopamine stored in the synaptic terminals
- d. It is used as conjunctive therapy for dyskinesia produced by levodopa

208. Which of the following agents would be preferred for the treatment of the drug-induced form of parkinsonism?

- a. Dopamine
- b. Levodopa
- c. Benztropine
- d. Bromocriptine

209. _____ is used in opioid withdrawal management because of its ability to block NMDA receptors and reduce cravings and withdrawal symptoms.

- a. Methadone
- b. Naloxone
- c. Naltrexone
- d. Buprenorphine

210. Which of the following antiparkinsonism drugs is an anticholinergic agent?

- a. Trihexyphenidyl
- b. Selegiline
- c. Bromocriptine
- d. Amantadine

211. Which of the following agents is the most helpful in counteracting the behavioral complications of levodopa?

- a. Carbidopa
- b. Clozapine
- c. Chlorpropamide
- d. Carbamazepine

212. Post traumatic brain injury administration of phenytoin is to prevent early onset post traumatic seizures within the first:

- a. 1 month
- b. 72 hours
- c. 24 hours
- d. 1 week

213. The actions of _____ CANNOT be described as that of a moderate opioid agonist.

- a. oxycodone
- b. hydromorphone
- c. fentanyl
- d. codeine

214. A house officer was loading a brain injury patient with IV phenytoin as a bolus but was asked by his boss to stop immediately and slowly infuse. The main reason for stopping the bolus infusion was:

- a. Phenytoin induced gingivitis
- b. Possible complication of brain swelling
- c. Phenytoin induced confusion
- d. Possible complication of arrhythmias

215. Regarding Bell's Palsy which of the following is TRUE?

- a. Routine use of antiviral therapy typically increases frequency of complete recovery
- b. An examination feature may be loss of taste to posterior 1/3 of tongue
- c. Less than 40% of those affected will have permanent paralysis
- d. Most common in 1st trimester compared to other trimesters of pregnancy

216. Principal aim for treatment of Parkinson's disease is to:

- a. restore dopaminergic activity with levodopa and dopamine agonists
- b. facilitate inhibitory neurotransmission in the CNS by enhancing the action of GABA
- c. decrease glutamatergic activity with glutamate antagonists
- d. restore the normal balance of cholinergic and dopaminergic influences on the basal ganglia with antimuscarinic drugs

RADIOLOGY

217. The non-contrast axial CT scan image of the brain post trauma is diagnostic of:

- a. Acute left temporal epidural hematoma
- b. Fracture of the left temporal bone
- c. Acute right frontal lobe hematoma
- d. Acute subarachnoid hemorrhage

218. Communicating hydrocephalus:

- a. Shows as hypo intense ventricular dilatation on T2 MRI
- b. Presents as hyper intense pan-ventricular dilatation on T2 weighted images
- c. Shows normal ventricles and gyri on both t2 and t1 weighted images
- d. Is not diagnosed prenatally via ultrasound

219. The following are CT scan features of early infarctive stroke except:

- a. Effacement of the sulci
- b. Hypodense lesion with CFS like density
- c. Insular ribbon sign
- d. Loss of grey white matter differentiation

220. Myelography is:

- a. A procedure for localizing brain lesions
- b. A non-contrast examination of the CNS
- c. Contrast procedure for examining the spinal cord
- d. It is routinely performed in the modern radiology centers

221. Ultrasonography of the brain is:

- a. Performed for an adult in diagnosing hydrocephalus
- b. A reliable source of diagnosing brain pathology in all age groups
- c. For diagnosing intracranial lesions like hydrocephalus in neonates and infants
- d. A radiation based examination not commonly used in children

222. Brain metastasis presents on CT scan as:

- a. Multiple hyper intense lesions of varying sizes and locations and associated perilesional hematoma
- b. Multiple hyper dense intracerebral lesions with massive perilesional oedema and may cause pressure effect
- c. Single cystic intracerebral mass
- d. Diffuse hypo dense intracerebral lesions without contrast enhancement and without oedema around the lesions

223. Acute subdural hematoma:

- a. Presents as a hyper biconvex lesion in acute phase on CT scan
- b. Presents as a hyper dense biconvex lesion in chronic stage on CT scan
- c. Presents as a non-enhancing biconvex hyper dense lesion in acute stage on CT scan
- d. Presents as a hyper dense crescentic lesion in the acute stage on CT scan

224. Acute Subarachnoid hemorrhage:

- a. Presents as iso dense lesions on CT scan

- b. Presents as hypo dense lesions on CT scan
- c. Presents as hyper densities in the external CSF spaces, quadrigeminal cisterns and in the ventricles on CT scan.
- d. Presents as non- enhancing hyper intense lesions on CT scan

225. A 60year old man present to the A and E with new onset seizures, headache and left hemiparesis. The radiologist report that, there is a nodular, ring enhancing lesion with a necrotic core in the right frontal lobe surrounded by hypodense vasogenic oedema and midline shift to the left. The most likely diagnosis if the patient has no fever is:

- a. Subacute infarct
- b. Abscess
- c. Glioblastoma multiforme
- d. Radiation necrosis

226. A non-contrast CT scan of a 67year old man shows dilatation of the lateral, third and fourth ventricles. No obstruction within the ventricular system. There is no corresponding dilatation of the cerebral cortical sulci. These findings are suggestive of:

- a. Intraventricular hemorrhage
- b. Age related brain atrophy
- c. Normal pressure hydrocephalus
- d. Obstructive hydrocephalus

227. A 45 year old male patient with sudden onset of right hemiparesis and motor aphasia presented to the accident and emergency department a moment ago. What imaging features below could explain his presenting clinical features:

- a. Right hemisphere infarct involving the motor cortex and Broca's area
- b. Left hemisphere infarct involving the motor cortex and Broca's area
- c. Left hemisphere infarct involving the motor cortex only
- d. Right hemisphere infarct involving the motor cortex only.

228. A 65-year-old right-handed man is brought in to the medical emergency. His wife reports that he has right upper and right lower limb weakness and unsteady walking that started suddenly. He is also reported to have difficulty speaking. Examination reveals normal in the left upper and lower limbs but power of 2/5 in the right upper and lower limbs. Which of the following computed tomography (CT) angiogram findings is most consistent with the patient's symptoms?

- a. Stenosis Of the left anterior cerebral artery
- b. Stenosis of the left middle cerebral artery
- c. Stenosis of the right posterior cerebral artery
- d. Stenosis Of the right vertebral artery

229. A 20 year old male patient with a one month history of seizures and chronic headaches has presented at the OPD. A plain head CT scan done shows an intracerebral haematoma. The investigations to determine the cause of the bleed would NOT include:

- a. Angiogram to rule out Arteriovenous malformation
- b. Test to rule out blood cancers
- c. Test to rule out pituitary tumours
- d. Test to rule out haemophilia

230. A 40 year old patient presented to the OPD with chronic headaches and seizures. Brain CT showed multiple small areas of calcification throughout the whole brain. This image is MOST likely suggestive of:

- a. Cerebral toxoplasmosis
- b. Neurohydatid cysts
- c. Neurocysticercosis
- d. Cerebral abscesses

231. A 10-year-old patient has been confirmed to have suffered a haemorrhagic stroke on imaging. Investigations for an underlying cause will NOT include:

- a. Angiogram of the brain

- b. Test for haemophilia
- c. Contrasted brain imaging
- d. Test for HIV

CHEMICAL PATHOLOGY

232. The daily production of CSF in humans:

- a. 150ml
- b. 155ml
- c. 350ml
- d. 550ml

233. The daily turnover of CSF in humans is:

- a. 3.3 times
- b. 3.5 times
- c. 3.7 times
- d. Unknown

234. Total volume of cerebrospinal fluid (CSF) in humans is:

- a. 350ml
- b. 550ml
- c. 155ml
- d. 150ml

235. Which of the following is NOT a primary function of cerebrospinal fluid (CSF)?

- a. Protection
- b. Nutrient transport

- c. Nerve impulse transmission
- d. Waste elimination

236. Which CSF parameter can help differentiate between bacterial and viral meningitis?

- a. Glucose concentration
- b. White blood cell count
- c. Protein concentration
- d. Red blood cell count

237. In the presence of a subarachnoid haemorrhage, which CSF parameter is likely to be increased?

- a. Lactate concentration
- b. Red blood cell count
- c. Total protein concentration
- d. Glucose concentration

238. Which enzyme is often used as a marker for neuronal damage when assessing CSF?

- a. Creatine kinase
- b. Glutamine synthetase
- c. Amylase
- d. Lipase

239. Elevated CSF protein levels can be indicative of:

- a. Inflammation or infection
- b. Hyperglycaemia
- c. Dehydration
- d. Hypertension

240. Which CSF component is measured to assess blood-brain barrier integrity?

- a. IgG
- b. Glucose
- c. Lactate
- d. Protein

241. Xanthochromia refers to:

- a. Decreased CSF glucose levels
- b. The presence of yellow-colored CSF
- c. Increased CSF protein levels
- d. Abnormal CSF pressure

242. In Major depressive disorder, early morning CSF findings of Low 5 HIAA suggests deficiency in:

- a. Serotonin
- b. Tryptophan
- c. Dopamine
- d. Noradrenaline

243. What indicates elevated levels of Tau in cerebrospinal fluid (CSF)?

- a. Progression of Alzheimer's disease.
- b. Disruption in normal processing of amyloid precursor protein.
- c. Genetic predisposition to Alzheimer's disease.
- d. Presence of A β plaques.

244. You have been asked to do a lumbar puncture on a 17 year old male patient with suspected meningitis. The following gross property of CSF is least important:

- a. Opening pressure of the CSF
- b. Osmolarity of the CSF
- c. Colour of CSF
- d. Density of the CSF

OTHERS

245. Spelling “world” backward in a mental status examination most closely tests which of the following?

- a. Memory
- b. Abstraction
- c. Executive function
- d. Attention

246. Which of the following is an appropriate question/command that forms part of a formal mini mental state examination?

- a. Spell the word ‘capture’.
- b. Where were you born?
- c. Three stage command: e.g. “Take this paper in your right hand, fold it in half, and put it on the table”
- d. What is your date of birth?

247. In the neurological examination, the myotome for wrist extension is:

- a. C3/4
- b. T1/T2
- c. C4/5
- d. C6/7

248. A patient has weakness in flexing her elbow joint. The root value of the nerve most likely affected is...

- a. C5, C6
- b. T1, T2
- c. C2, C3
- d. C3, C4

249. On examination of her lower limbs a patient is only able to feeble contractions that are unable to move a joint in her right lower limb whilst she is able to lift her left lower limb off the bed unaided. The power in her right and left lower limbs are respectively...

- a. 0 and 4
- b. 1 and 2
- c. 0 and 3
- d. 1 and 3

250. Which of the following clinical features suggests a delirium more so than psychosis:

- a. Delusions
- b. Auditory hallucinations
- c. Normal attention
- d. Fluctuating conscious state

251. What is the main goal of therapy for a patient being treated for Alzheimer's disease?

- a. Effectively treat psychiatric and behavioural complications
- b. Manage financial costs of the patient's therapy
- c. Maintain the patient's independence
- d. Reverse the patient's cognitive decline

252. The symptoms of the lower motor neuron lesion are the following:

- a. muscular spasticity, muscular flaccidity
- b. deep reflexes depressed, muscular wasting, EMG signs
- c. Babinsky response positive, hyperreflexia
- d. intention tremor, ataxia

253. The upper motor neuron lesion signs are the following:

- a. fibrillations, muscular flaccidity
- b. muscular spasticity, deep reflexes brisk, depressed superficial reflexes, Babinsky response positive
- c. hypotonia, areflexia
- d. intention tremor, nystagmus

254. The following symptoms are the most common in cauda equine affection:

- a. burning ache in legs, anesthesia in lower extremities & perineum, bladder dysfunction, flaccid paraplegia in legs
- b. spastic paraplegia, hyperreflexia
- c. anesthesia below nipples, hypertonia
- d. Horner's syndrome, ataxia

255. Midbrain affection may include the following symptoms:

- a. oculomotor nerve palsy, diplopia, ptosis, hemiplegia
- b. dysphagia, atrophy of the tongue
- c. ataxia, dysmetria
- d. anosmia, visual field defects

256. The following symptoms are typical for bulbar palsy:

- a. diplopia, ptosis
- b. atrophy of the tongue, dysphagia, dysarthria, dysphonia

- c. hemiplegia, hemianesthesia
- d. ataxia, dysmetria

257. The typical signs of the facial nerve damage are the following:

- a. dysphagia, paresis of the tongue
- b. lagophthalmus, Bell's sign, inability to whistle, depressed eyebrow reflex
- c. ptosis, miosis, anhidrosis
- d. visual field defects, papilloedema

258. The following symptoms are the most common in trigeminal ganglion impairment:

- a. sensory disturbance in the face & herpes rash, facial ache, depressed corneal reflex
- b. weakness in facial muscles, depressed eyebrow reflex
- c. disturbance of mastication, dysphagia
- d. ptosis, diplopia

259. The following symptoms may occur in optic nerve damage:

- a. blindness, scotoma, depressed pupil reaction
- b. exophthalmus, ptosis, diplopia
- c. visual field defects, papilloedema
- d. nystagmus, ataxia

260. Symptoms typical for cerebellar affection are the following:

- a. gait ataxia, muscular hypotension, dyscoordination in extremities, nystagmus
- b. dysphasia, paresis in extremities
- c. hyperreflexia, spasticity
- d. chorea, athetosis

261. The following symptoms are the most typical for polyneuropathy:

- a. sensory loss in dermatomes, hemianesthesia
- b. pain & numbness in extremities, anesthesia in distal parts of extremities, depressed ankle jerks
- c. Babinsky response positive, hyperreflexia
- d. ataxia, dysmetria

262. Hemianopia may develop with damage in:

- a. chiasma opticum, optic tract, occipital lobe, thalamus opticus
- b. optic nerve, frontal lobe
- c. cerebellar cortex, pons varolii
- d. medulla, hypoglossal nerve

263. Dysphasia may develop with damage in:

- a. cerebellar cortex, pons varolii
- b. frontal lobe cortex, temporal lobe cortex
- c. medulla, hypoglossal nerve
- d. thalamus, basal ganglia

264. The higher cortical (cognitive) dysfunction is the following:

- a. dysphasia, dyslexia, dysmnnesia
- b. dysphonia, dysdiadochokinesis, dysmetria
- c. ataxia, chorea
- d. spasticity, rigidity

265. The following symptoms may develop with damage in hypothalamus:

- a. face paresis, hemianalgesia
- b. panic attacks, sweating disorder, blood pressure disturbance, cardiac rhythm disturbance

- c. ataxia, dysmetria
- d. dysphasia, dyslexia

266. The signs of intracranial hypertension are the following:

- a. loss of hearing, panic attack
- b. headache in the morning, nausea & vomiting, bradycardia, papilledema
- c. ataxia, nystagmus
- d. hemiparesis, ptosis

267. The general (universal) brain symptoms are the following:

- a. hemiparesis, ptosis, dysphasia
- b. headache, loss of consciousness, vomiting
- c. ataxia, dysmetria
- d. chorea, athetosis

268. The focal neurological symptoms are the following:

- a. headache, vomiting, alteration of consciousness
- b. hemiparesis, ptosis, disturbance of coordination & balance
- c. fever, neck stiffness
- d. papilledema, bradycardia

269. Meningeal symptoms are the following:

- a. Lasegue's sign, Nery's sign, Babinsky sign
- b. Kernig's sign, neck stiffness, Brudzinsky sign
- c. ataxia, dysmetria
- d. chorea, athetosis

270. Symptoms typical for Parkinsonism are the following:

- a. muscular flaccidity, intention tremor
- b. rest tremor, muscular “cog wheel” rigidity, “mask” face, bradykinesia
- c. hyperreflexia, spasticity
- d. ataxia, nystagmus

271. Involuntary movements (hyperkinesia) are the following syndromes:

- a. bradykinesia, “cog wheel” rigidity, paresis
- b. chorea, torsion dystonia, atetosis
- c. ataxia, dysmetria
- d. spasticity, flaccidity

272. A 34-year-old man is brought in to the hospital with altered consciousness. He opens his eyes briefly when his name is called. He is heard muttering a mixture of words which don't make any sense and his limbs flex abnormally in reaction to knuckles being rubbed on his sternum. What is his score on the Glasgow Coma Scale?

- a. 7
- b. 8
- c. 9
- d. 10

273. A 40-year-old patient with no known co-morbidities presents to the Accident and Emergency department with a sudden severe headache followed by a reduced level of consciousness. Initial diagnostic assessment of this patient DOES NOT include:

- a. CT scan of the chest
- b. CT scan of the brain
- c. Estimation of Glasgow coma scale score
- d. Estimation of pupil sizes

274. In taking a myelomeningocele patient's history one of the most important pieces of information from the pregnancy history is enquiring about dietary:

- a. Follicle stimulating hormone
- b. Essential amino acids
- c. Folate
- d. Vitamin C

275. Clinical features of patients that raise the suspicion of a hidden spinal dysraphism include:

- a. birth mark over the lumbar area
- b. Tuft of hair over the lumbar area
- c. Prominent vein over the back
- d. Pimple over the lumbar area

276. Undue delay in the management of pituitary apoplexy in a 40-year-old woman led to blindness in the both eyes. What is the mechanism behind this complication?

- a. Sudden increase in intrasellar pressure which compresses cranial nerve II and the optic chiasm
- b. Sudden bleed into the pituitary tumour with disruption of hormone production
- c. Infarction usually affects the anterior pituitary which swells and compress the posterior part
- d. Sudden increase in intrasellar pressure which compresses the optic chiasm and cranial nerve III

277. A 20 years old male patient has just been diagnosed with a contre-coup brain injury after a brain CT scan. Which of the following injury mechanisms can result in his condition?

- a. Stabbed in the head with a knife
- b. Gun shot from the left to the right side of the head
- c. Hit on the head with a hammer
- d. Fall from a coconut tree

278. A 15-year-old female patient knocked down by a motor bike has base of skull fracture after imaging: Which of the following signs is LEAST likely to be found in her?

- a. Facial nerve palsy
- b. Battle sign
- c. Racoon eyes
- d. Optic nerve atrophy

279. Which of the follow best describes vertigo due to benign paroxysmal positional vertigo?

- a. Episodic of >30 minutes and lasts up to 24 hours, no recent infections, horizontal nystagmus, unilateral hearing loss, associated tinnitus
- b. Gradual onset with persistent imbalance, no nystagmus, associated unilateral hearing loss but no vertigo
- c. Gradual onset lasting days, not associated with head movements, horizontal nystagmus at rest, no associated hearing loss
- d. Episodic lasting up to 1 minute, associated with head movements, rotational nystagmus when head turned, no hearing loss

280. A 20-year-old male patient with a large tumour obliterating the 4th ventricle is in need of a lifesaving procedure before tumour resection. Which of the following is contraindicated?

- a. External ventricular drain
- b. Endoscopic third ventriculostomy
- c. Lumoperitoneal shunt
- d. Ventriculoperitoneal shunt

281. The following is NOT a likely differential diagnosis of a patient presenting typically with photophobia, headache and neck stiffness:

- a. Subdural empyema
- b. Chemical meningitis
- c. Bacterial meningitis

d. Subarachnoid haemorrhage

282. The following is NOT included in the Nursing instructions you would give to optimize the management of a patient with confirmed subarachnoid haemorrhage:

- a. Administer clexane: 10mg daily
- b. Limit disturbance from visitors
- c. Apply TED stockings
- d. Regularly turn patient to prevent bed sores

283. A CT scan of a 75year old man shows an irregular hyperdense collection with surrounding oedema in the right posterior frontal and parietal lobes with midline shift to the left. His BP on arrival is 280/110mmHg. The CT scan findings are most likely consistent with:

- a. Acute right fronto-parietal lobe hemorrhage
- b. Acute right fronto-parietal subdural hemorrhage
- c. Acute right fronto-parietal epidural hemorrhage
- d. Acute right fronto-parietal subarachnoid hemorrhage

284. Which of the following conditions can secondarily cause brain injury in a patient with severe diffuse brain injury?

- a. Hypocapnia and normotension
- b. Normalized blood pressure and hypoxia
- c. Hypoxia and fever
- d. Hypoxia and normocapnia

285. A 60year old man present to the A and E with new onset seizures, headache and left hemiparesis. The first modality to investigate this patient is:

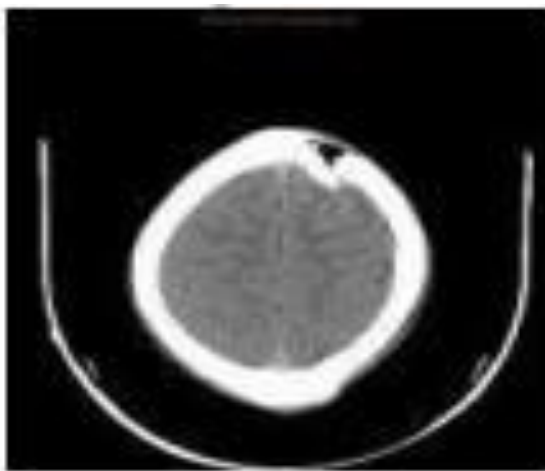
- a. Non-contrast CT scan of the brain
- b. Skull radiograph
- c. MRI of the brain

d. Transcranial Ultrasound

286. Which of the following patients with diffuse brain injury is deemed to have an absolute indication for intubation and ventilation? The Patient with GCS:

- a. 1/5-8/15
- b. 3/15-14/15
- c. 3/15-8/15
- d. 3/15-10/15

287. The image below is that of a 23 year old female patient hit with a wooden plank. What condition does she have?



- a. Open depressed skull fracture
- b. Acute extradural haematoma
- c. Acute subdural haematoma
- d. Depressed skull base fracture

288. A 20 year old male patient with a one month history of seizures and chronic headaches has presented at the OPD. A plain head CT scan done shows an intracerebral haematoma. The investigations to determine the cause of the bleed would NOT include:

- a. Angiogram to rule out Arteriovenous malformation

- b. Test to rule out blood cancers
- c. Test to rule out pituitary tumours
- d. Test to rule out haemophilia

289. The scan below is that for a 14-year-old male patient who was stabbed in the head 2 weeks ago. What condition is depicted in the scan:

- a. Subdural empyema
- b. Cerebral haematoma
- c. Subgaleal pus collection
- d. Brain abscess

290. The following procedure can be done to correct the condition shown below:

- a. Syringoparitoneal shunt
- b. Endoscopic fourth ventriculostomy
- c. Endoscopic third ventriculostomy
- d. Syringoatrial shunt

291. A 4-year-old patient known with congenital hydrocephalus and a VP shunt in situ now presents with a 3 day history of headaches followed by reduced level of consciousness, no fever. What is the most likely cause of his condition?

- a. VP shunt infection
- b. VP shunt blockage
- c. Intestinal obstruction from adhesions
- d. Calcification of the ventricular catheter

292. Which of the following abnormalities would NOT be expected in a severe, ulnar neuropathy?

- a. brachioradialis reflexes are decreased

- b. Positive Froment sign
- c. numbness in the fifth digit
- d. claw hand

293. A 67-year-old woman said she was unable to use utensils to eat her food or use her keys. She says this problem began upon awakening. On exam, there were no abnormalities of strength, sensation, or coordination. Which of the following is the term for this disorder?

- a. Malingering
- b. Prosopagnosia
- c. Apraxia
- d. Astereognosia

294. A 60-year-old known hypertensive with poorly controlled BPs has just developed a weakness on the right with a left dilated pupil. Which of the following imaging pictures is likely to be depicted:

- a. Left basal ganglia bleed
- b. Right basal ganglia bleed
- c. Right cerebellar bleed
- d. Left cerebellar bleed

295. On examination of her lower limbs a patient is only able to feeble contractions that are unable to move a joint in her right lower limb whilst she is able to lift her left lower limb off the bed unaided. The power in her right and left lower limbs are respectively...

- a. 1 and 3
- b. 1 and 2
- c. 0 and 3
- d. 0 and 4

296. What is the role of Tau protein in neurons?

- a. It forms plaques in Alzheimer's disease.
- b. It is involved in lipid metabolism.
- c. It stabilizes neuronal axons.
- d. It is released into the bloodstream upon neuronal injury.

297. Regarding Guillain-Barré syndrome which of the following is INCORRECT?

- a. CSF may only show raised protein levels
- b. The mainstay of treatment is corticosteroids
- c. Campylobacter infections is the chief precipitant in up to 25% of cases
- d. Miller-Fischer variant has predominantly cranial nerve involvement

298. Which of the following descriptions best fits the diagnosis of a Holmes-Adie pupillary disorder?

- a. Large pupil that does not react to light or accommodation
- b. Small pupils that react briskly to light and accommodation
- c. Large pupil that reacts slowly to light and accommodation
- d. Small irregular pupils that accommodate but do not react to light

299. A mother complains to you that her 8yr old boy has started walking like a rogue and will not stop even when cautioned. She says he has seen him fall because of that funny walk and now he has started complaining of headache. A night before bringing him to the emergency he started convulsing. Your direct questioning reveals that he is afebrile. Your examination further reveals the child has balance and gait abnormalities and may actually be having signs of increased intracranial pressure from an intracranial tumour. A CT scan suggests that the child has an increase in the volume of CSF, this is most likely a:

- a. None of the other options
- b. Non communicating hydrocephalus
- c. Communicating hydrocephalus
- d. Hydrocephalus ex-vacuo

300. An 80-year-old man sees you for slurred speech, which is evident on office examination. Notable findings include deviation of his tongue to the left, when asked to stick his tongue out straight. The left side of the tongue looks withered and “twitchy.” The rest of his neurological examination is normal. Where is the most likely lesion here?

- a. left glossopharyngeal nerve
- b. left hypoglossal nerve
- c. left facial nerve
- d. right frontal motor cortex

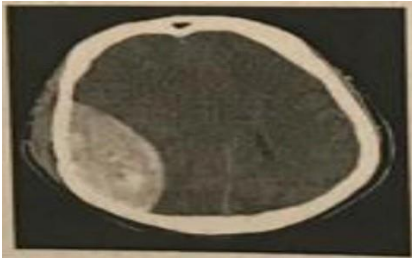
301. A 45 year old man presented to the Accident and Emergency Unit with a sudden severe lower back pain which radiates into the sole of the right foot after lifting a heavy object. The most likely diagnosis is:

- a. Brown-Sequard syndrome
- b. Cauda Equina syndrome
- c. Right Lumbar myelopathy
- d. Right S1 radiculopathy

302. A former rugby player presents to the OPD with a two year history of gradually worsening spastic weakness of both arms and legs. He is now unable to do the buttons Of his shirt and walks slowly with a spastic gait. What is the most likely diagnosis?

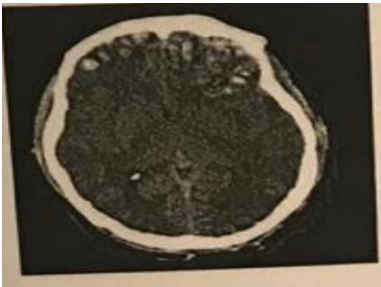
- a. Cervical myelopathy
- b. Cervical radiculopathy
- c. Lumbar myelopathy
- d. Neurogenic claudication

303. A 20 year old student playing soccer suffers a head collision on the field of play followed by severe headache and vomiting. CT scan done is as shown below. What injury did he sustain?



- a. Extradural hematoma
- b. Extramural hematoma
- c. Subdural hematoma
- d. Intracerebral hematoma

304. The CT Brain scan of a 45 year Old male patient involved in a road traffic accident is as shown below. The injury shown in the scan is:



- a. Intraventricular hematoma
- b. Multiple extradural hematomas
- c. Multiple contusions
- d. Multiple subdural hematomas

305. A 30 year old road traffic accident victim is found to have a left hemiplegia and a right fixed dilated pupil. CT Brain scan done showed a brain injury. Which of the following injuries is most likely to be depicted on the scan?

- a. Bilateral subarachnoid hemorrhage
- b. Left subdural hematoma
- c. Multiple contusions

d. Right subdural hematoma

306. A 45 year old known hypertensive presented to Medical Emergency with a history Of sudden severe headaches followed by right arm and leg weakness and aphasia. CT Brain done is as shown below.



- a. aspirin and use of anticoagulants to prevent DVT
- b. Control and optimization Of her blood pressure
- c. Head up 300, neck in neutral position with minimal patient stimulation
- d. Prompt involvement of speech therapy and physiotherapy

307. A 35 year old woman presents to the emergency department with a history of spontaneous sudden severe headaches, neck stiffness , photophobia and cranial nerve III palsy on the right. The most likely cause of this presentation is:

- a. SAH from ruptured cerebral aneurysm
- b. SAH from trauma to the head
- c. SAH from ruptured preretinal arteries
- d. Subdural hematoma from ruptured bridging veins

308. A ct brain image shown in that of a patient hit on the head with a hockey stick. What type of skull fracture did he sustain?

- a. Linear fracture
- b. Stellate fracture
- c. Depressed skull fracture

d. Base of skull fracture

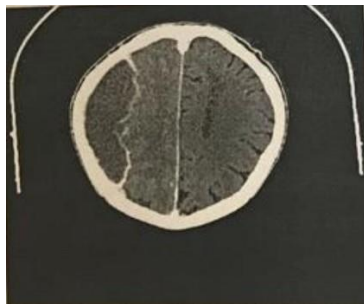
309. A road traffic accident victim with GCS 15/15, no focal deficit was found the next morning at the during neurosurgical ward rounds to be lying prone with a bull's eye sign (thick rounded stain in the middle with a light clear fluid stain around) on the white linen of his pillow case. The bull's eye sign is indicative of:

- a. A very severe brain injury
- b. CSF rhinorrhea from base Of skull fracture
- c. Fluid leak on a fractured nose
- d. Fluid leak from a ruptured eye ball

310. A 40 year old female trader who reported to the Neurosurgery clinic with a CT brain was reported to have a left sphenoid ridge meningioma with surrounding oedema and midline shift to the right. The type of oedema reported above is most likely to be:

- a. Vasogenic oedema
- b. Vasospastic oedema
- c. Vasodelic oedema
- d. Vasospasmic oedema

311. A 24 year old lady reported to the Maxillofacial clinic with a dental abscess and a progressive left sided arm and leg weakness, Her post contrasted CT brain scan shown below is that of:



- a. an extradural abscess
- b. an intracerebral abscess

- c. subdural empyema
- d. intraventricular empyema

312. 13 month old baby was sent to the clinic due to an enlarging head circumference and a bulging anterior fontanelle with a shrill cry. Ct scan reveals a communicating hydrocephalus. Which of the ff is not a cause of communicating hydrocephalus?

- a. Arnold-chiari malformation
- b. dural sinus thrombosis
- c. gliosis
- d. organized meningitis

313. A 60 year old male is seen on admission with a ct scan report stating he had a brain tumor located in the left frontal lobe. The patient creatinine is 61 $\mu\text{mol/l}$, Na 128, Cl 4.0mmol/l, urine osmolality is raised. Which of the ff clinical features will correspond to the site of the patient lesion?

- a. Apraxia
- b. Loss of two-point discrimination
- c. Acalculia
- d. Loss of executive function

314. Which of the ff clinical features would not be explained by the site of patient lesion?

- a. Loss of judgement
- b. Loss of stereognosis
- c. Disinhibition
- d. Loss of attention

315. Which complication of the patient condition has developed?

- a. SIADH
- b. Metastasis to the kidney

- c. Secondary hypocortisolism
- d. Dehydration

316. A 58 year old female patient was brought to the clinic by her husband who complained that she had gradually lost the normal range of motion in her right upper limb with no swinging of that limb when she walked and stiffness in the arm that started a year ago. More recently she began to drag her right leg. Her handwriting has become very small and difficult to read and she has a resting tremor in her right arm. Her head ct that was ordered was reported as being normal. Which of the ff conditions will most likely explain her symptoms?

- a. Stroke
- b. Parkinson's
- c. Epilepsy
- d. Syringomyelia

317. Which of the ff signs will be consistent with the most likely diagnosis in this patient?

- a. Asterixis
- b. Hypotonia
- c. Bradykinesia
- d. Disdiadokinesia

318. Which of the ff treatment will be most useful in this patient?

- a. Clopidogrel
- b. Pramipexole
- c. Valproate
- d. Carbamazepine

319. A 64 year old man presented with distal paresthesia in his hands and feet, an unsteady gait and frequent falls. His wife who accompanied him complained that she had noticed he also had difficulty remembering this. On examination he had power of 4/5 in his lower limbs, a broad based gait. Romberg's test was positive. He had lost position sense in his toes. Fbc revealed a hb

of 9g/dl, mcv 112fl, wbc $4 \times 10^9/l$, platelets of $120 \times 10^0/l$ ct myelogram was normal. Which of the ff clinical features would be consistent with the most likely diagnosis in this patient?

- a. Decreased tone in the lower limbs
- b. Areflexia
- c. Loss of vibration sense at the medial malleoli
- d. Muscle wasting

320. Which of the ff is a risk factor for developing this patient's condition?

- a. Previous gastrectomy
- b. Obesity
- c. Hypertension
- d. Recent campylobacter jejuni infection

321. Which of the ff would be used in the treatment of this patient?

- a. Aspirin
- b. Atorvastatin
- c. Nifedipine
- d. Vit b12

322. Which of the following conditions can secondarily cause brain injury in a patient with severe diffuse brain injury?

- a. Hypocapnia and normotension
- b. Normalized blood pressure and hypoxia
- c. Hypoxia and fever
- d. Hypoxia and normocapnia

323. A 60year old man present to the A and E with new onset seizures, headache and left hemiparesis. The imaging modality of choice is:

- a. Skull radiograph
- b. MRI of the brain
- c. Transcranial Ultrasound
- d. Non-contrast CT scan of the brain

324. A mother complains to you that her 8yr old boy has started walking like a rogue and will not stop even when cautioned. She says he has seen him fall because of that funny walk and now he has started complaining of headache. A night before bringing him to the emergency he started convulsing. Your direct questioning reveals that he is afebrile. Your examination further reveals the child has balance and gait abnormalities and may actually be having signs of increased intracranial pressure from an intracranial tumour. A glioma in this age group is likely to be an:

- a. Undifferentiated tumour
- b. Oligodendrogliomas
- c. Ependymoma
- d. Pilocytic astrocytoma

325. Difficulty identifying tactile sensations of written patterns (shapes or letters) on the skin is known as ...

- a. Simultagnosia
- b. Stereognosis
- c. Graphaesthesia
- d. Praxis

326. The above image is the post contrast axial MRI of A 16-year-old girl who presents with lethargy, headache and vomiting. The findings are highly suggestive of:

- a. Pilocytic astrocytoma
- b. Ependymoma
- c. Hemangioblastoma
- d. Medulloblastoma

327. A 34-year-old man is brought in to the hospital with altered consciousness. He opens his eyes briefly when his name is called. He is heard muttering a mixture of words which don't make any sense and his limbs flex abnormally in reaction to knuckles being rubbed on his sternum.

What is his score on the Glasgow Coma Scale?

- a. 7
- b. 8
- c. 9
- d. 10

328. A mother complains to you that her 8yr old boy has started walking like a rogue and will not stop even when cautioned. She says he has seen him fall because of that funny walk and now he has started complaining of headache. A night before bringing him to the emergency he started convulsing. Your direct questioning reveals that he is afebrile. Your examination further reveals the child has balance and gait abnormalities and may actually be having signs of increased intracranial pressure from an intracranial tumour. Another 4th year student is fascinated by the word 'glioma' and wishes to know what it means. You will explain that these tumours arise from:

- a. Meningothelial cells, ependymal cells, astrocytes and oligodendrocytes
- b. Macroglial cells and microglial cells
- c. Ependymal cells, astrocytes and microglial cells
- d. Ependymal cells, astrocytes and oligodendrocytes

329. Which of the following abnormalities would NOT be expected in a severe, ulnar neuropathy?

- a. brachioradialis reflexes are decreased
- b. Positive Froment sign
- c. numbness in the fifth digit
- d. claw hand

330. A mother complains to you that her 8yr old boy has started walking like a rogue and will not stop even when cautioned. She says he has seen him fall because of that funny walk and now he

has started complaining of headache. A night before bringing him to the emergency he started convulsing. Your direct questioning reveals that he is afebrile. Your examination further reveals the child has balance and gait abnormalities and may actually be having signs of increased intracranial pressure from an intracranial tumour. Supposing this is a glioma in this age group it is most likely to be:

- a. Spinal
- b. Infra-tentorial
- c. Supra-tentorial
- d. Subdural

331. The brain is ----- of body mass yet uses ----- of the daily glucose requirement of the body:

- a. 10%, 20%
- b. 20%, 10%
- c. 5%, 20%
- d. 15%, 15%

332. In Major Depressive Disorder the morning cerebrospinal fluid (CSF) levels of 5-Hydroxyindoleacetic acid (5-HIAA) if low indicates a deficiency in:

- a. serotonin.
- b. 5-HIAA.
- c. tryptophan.
- d. melanin.

333. A 40 year old patient with a 15 years smoking history has presented to the OPD with a 3 day history of sudden onset drooping of the right upper eye brow and a fully dilated right pupil with mild headaches. The most likely differential diagnosis is:

- a. Viral infection of the right eye.
- b. Graces disease with unilateral right eye effect
- c. Cranial nerve II palsy from cerebral aneurysm

d. Cranial nerve III palsy from cerebral aneurysm

334. Principal aim for treatment of Parkinson's disease is to:

- a. restore dopaminergic activity with levodopa and dopamine agonists
- b. facilitate inhibitory neurotransmission in the CNS by enhancing the action of GABA
- c. decrease glutamatergic activity with glutamate antagonists
- d. restore the normal balance of cholinergic and dopaminergic influences on the basal ganglia with antimuscarinic drugs

335. On examination of her lower limbs a patient is only able to feeble contractions that are unable to move a joint in her right lower limb whilst she is able to lift her left lower limb off the bed unaided. The power in her right and left lower limbs are respectively...

- a. 0 and 4
- b. 1 and 2
- c. 0 and 3
- d. 1 and 3

336. A 26-year-old male patient presents with a power of 3 in his lower limbs, depressed reflexes and muscle wasting. Which of the following features would put the site of the lesion at the level of the muscle?

- a. Predominantly proximal muscle weakness
- b. Abnormal nerve conduction test results
- c. The presence of fasciculation
- d. Muscle percussion-induced contraction

337. Undue delay in the management of pituitary apoplexy in a 40-year-old woman led to blindness in the both eyes. What is the mechanism behind this complication?

- a. Sudden increase in intrasellar pressure which compresses cranial nerve II and the optic chiasm
- b. Sudden bleed into the pituitary tumour with disruption of hormone production

- c. Infarction usually affects the anterior pituitary which swells and compress the posterior part
- d. Sudden increase in intrasellar pressure which compresses the optic chiasm and cranial nerve III

338. A 65-year-old right-handed man is brought in to the medical emergency. His wife reports that he has right upper and right lower limb weakness and unsteady walking that started suddenly. He is also reported to have difficulty speaking. Examination reveals an irregular pulse, normal power in the left upper and lower limbs but power of 2/5 in the right upper and lower limbs. Which of the following computed tomography (CT) angiogram findings is most consistent with the patient's symptoms?

- a. Stenosis of the left anterior cerebral artery
- b. Stenosis of the right vertebral artery
- c. Stenosis of the left middle cerebral artery
- d. Stenosis of the right posterior cerebral artery

339. The following factor is NOT a cause of secondary brain injury in a road traffic accident victim:

- a. Persistently low blood pressures
- b. Prolonged uncontrolled seizures
- c. Hitting of patient's head against the steering wheel
- d. Persistently low arterial oxygen saturation

340. A patient with very high fevers just after a severe post traumatic diffuse brain injury is most likely caused by:

- a. Disregulation of the patients' autonomic nervous system
- b. Disregulation of the patients entire nervous system
- c. Disregulation of the body's temperature
- d. Total shutdown of the patients immune system and subsequent infection

341. A 20 years old male patient has just been diagnosed with a contre-coup brain injury after a brain CT scan. Which of the following injury mechanisms can result in his condition?

- a. Stabbed in the head with a knife
- b. Gun shot from the left to the right side of the head
- c. Hit on the head with a hammer
- d. Fall from a coconut tree

342. Which of the following is true about Neurofilament Light Chains (NfL)?

- a. They stabilize neuronal axons.
- b. They are structural proteins found in neurons.
- c. They are only found in the bloodstream.
- d. Elevated levels are specific to Alzheimer's disease.

343. A 15-year-old female patient knocked down by a motor bike has base of skull fracture after imaging: Which of the following signs is LEAST likely to be found in her:

- a. Facial nerve palsy
- b. Battle sign
- c. Racoon eyes
- d. Optic nerve atrophy

344. Which of the following patients is most likely suffering from headaches due to temporal arteritis?

- a. 25-year-old female. Pain is bilateral, constant, no nausea, no neurological deficits. Relieved by NSAIDs.
- b. 75-year-old male. Pain throbbing and over frontotemporal region, history of polymyalgia rheumatica.
- c. 40-year-old male. Pain is sharp, and occipital in region, associated with vertigo and ataxia.
- d. 45-year-old female. Pain is sudden onset and global, had a loss of consciousness at the time, vomiting, no neurological deficit.

345. A mother complains to you that her 8yr old boy has started walking like a rogue and will not stop even when cautioned. She says he has seen him fall because of that funny walk and now he has started complaining of headache. A night before bringing him to the emergency he started convulsing. Your direct questioning reveals that he is afebrile. Your examination further reveals the child has balance and gait abnormalities and may actually be having signs of increased intracranial pressure from an intracranial tumour. You suspect that this child has increased intracranial pressure, your biggest fear is that the brain will herniate. This is most likely to be a:

- a. Tonsillar herniation
- b. Subfalcine herniation
- c. Mesial temporal herniation
- d. Any of the other options

346. In which of these patients is the diagnosis of Idiopathic intracranial hypertension most likely correct?

- a. 60-year-old male. Pain is diffuse, rapid onset, associated with neck stiffness, photophobia. Afebrile in ED but did describe shivering at home.
- b. 65-year-old male. Pain dull over forehead, worsened when leaning forward, recent URTI symptoms, fever at 38 degrees.
- c. 35-year-old female. Pain is diffuse, gradual onset, associated with vomits and started few days after birth of 1st baby
- d. 30-year-old female. Pain has been ongoing for weeks, constant and worsened by coughing, associated nausea and blurred vision, BMI 34

347. The image below shows:



- a. Chronic right insular cortex infarct
- b. Acute left insular cortex infarct
- c. Chronic left insular cortex infarct
- d. Acute right insular cortex infarct

348. A house officer was loading a brain injury patient with IV phenytoin as a bolus but was asked by his boss to stop immediately and slowly infuse. The main reason for stopping the bolus infusion was:

- a. Phenytoin induced gingivitis
- b. Possible complication of brain swelling
- c. Phenytoin induced confusion
- d. Possible complication of arrhythmias

349. Which of the following conditions can secondarily cause brain injury in a patient with severe diffuse brain injury?

- a. Hypocapnia and normotension
- b. Normalized blood pressure and hypoxia

- c. Hypoxia and fever
- d. Hypoxia and normocapnia

350. Which of the following secondary symptoms is most frequently caused by idiopathic intracranial hypertension?

- a. Conductive hearing loss
- b. Pulsatile tinnitus
- c. Sensorineural hearing loss
- d. Palatal myoclonus

351. A 67-year-old man was involved in high impact accident. Complains of neck pains on arrival at the emergency room. A CT scan of the cervical spine shows a fracture through the base of the dens. This is classified as:

- a. Type I
- b. Type II
- c. Type III
- d. Type IV

352. The scan below is that for a 14-year-old male patient who was stabbed in the head 2 weeks ago. What condition is depicted in the scan:

- a. Subdural empyema
- b. Cerebral haematoma
- c. Subgaleal pus collection
- d. Brain abscess

353. A mother complains to you that her 8yr old boy has started walking like a rogue and will not stop even when cautioned. She says he has seen him fall because of that funny walk and now he has started complaining of headache. A night before bringing him to the emergency he started convulsing. Your direct questioning reveals that he is afebrile. Your examination further reveals the child has balance and gait abnormalities and may actually be having signs of increased

intracranial pressure from an intracranial tumour. The neurosurgeon enucleates the lesion and sends it for histopathological analysis. The pathologist describes a lesion that has chronic granulomatous inflammation with caseous necrosis. His diagnosis will likely be:

- a. A meningioma
- b. An abscess
- c. A tuberculoma
- d. A glioma

354. A 60-year-old known hypertensive with poorly controlled BPs has just developed a weakness on the right with a left dilated pupil. Which of the following imaging pictures is likely to be depicted:

- a. Left basal ganglia bleed
- b. Right basal ganglia bleed
- c. Right cerebellar bleed
- d. Left cerebellar bleed

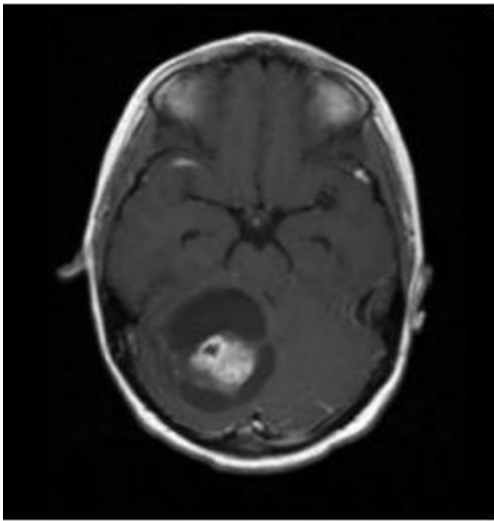
355. A mother complains to you that her 8yr old boy has started walking like a rogue and will not stop even when cautioned. She says he has seen him fall because of that funny walk and now he has started complaining of headache. A night before bringing him to the emergency he started convulsing. Your direct questioning reveals that he is afebrile. Your examination further reveals the child has balance and gait abnormalities and may actually be having signs of increased intracranial pressure from an intracranial tumour. A CT scan suggests that the child has an increase in the volume of CSF, this is most likely a:

- a. None of the other options
- b. Non communicating hydrocephalus
- c. Communicating hydrocephalus
- d. Hydrocephalus ex-vacuo

356. A 40-year-old patient with no known co-morbidities presents to the Accident and Emergency department with a sudden severe headache followed by a reduced level of consciousness. Initial diagnostic assessment of this patient DOES NOT include:

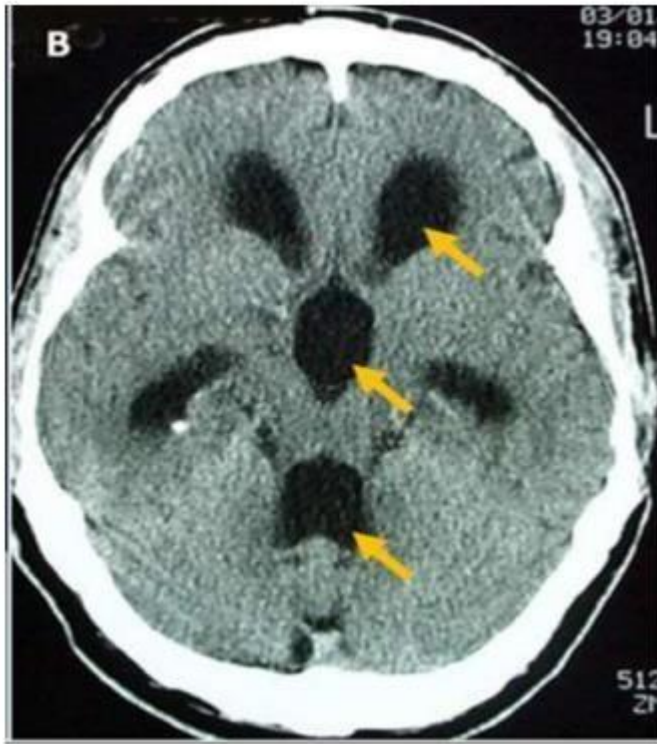
- a. CT scan of the chest
- b. CT scan of the brain
- c. Estimation of Glasgow coma scale score
- d. Estimation of pupil sizes

357. The image above depicts axial non contrast cervical spine CT scan, bone window, of a patient with suspected neck injury. The likely diagnosis is



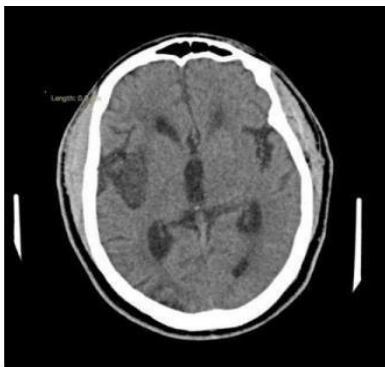
- a. Type 3 = Fracture of both the anterior and posterior arches
- b. Type 1 = Fracture of the posterior arch only
- c. Type 4 = Fracture of the lateral mass(es) of C1
- d. Type 2 = Fracture of the anterior arch only

358. The non-contrast CT scan image above shows



- a. Age related brain atrophy
- b. Obstructive hydrocephalus
- c. Intraventricular hemorrhage
- d. Communicating hydrocephalus

359. The significant feature of the image above is?



- a. Subacute left subdural hematoma
- b. Chronic left subdural hematoma
- c. Acute left subdural hematoma
- d. Intraparenchymal hyperdensities

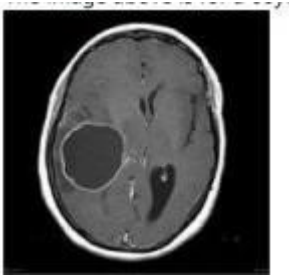
360. The non-contrast CT scan image below of 75y old man with hypertension



- a. Acute right frontal lobe hematoma
- b. Acute Subarachnoid hematoma
- c. Acute subdural hematoma
- d. Acute epidural hematoma

361. The image above is for a 60year old man with new onset seizure and fever.

362. The likely diagnosis is



- a. Subacute infarct
- b. Glioblastoma multiforme
- c. Cerebral abscess
- d. Radiation necrosis

363. The non-contrast axial CT scan image of the brain post trauma is diagnostic of



- a. Acute subarachnoid hemorrhage
- b. Acute right temporal lobe hematoma with contrecoup left temporal lobe hematoma
- c. Fracture of the left temporal bone
- d. Acute left temporal epidural hematoma with contrecoup right temporal lobe hematoma

364. The significant feature of the image below is?



- a. Widening of the sulci
- b. Intraparenchymal hyperdensities
- c. Hyperdensities within the sulci
- d. Subacute left subdural hematoma

~~Handwritten signature~~
Kall